Action-Conditioned Predictive Consistency as a Diagnostic for Gaussian-Noise Robustness in JEPA World Models

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June 2026

Abstract

Latent predictive world models such as JEPAs avoid pixel reconstruction, but latent prediction alone does not define robustness for closed-loop control. We study additive Gaussian pixel noise as the primary controlled diagnostic stressor for JEPA world-model control. We formalize action-conditioned predictive consistency (ACPC): clean and noised views of the same state should induce similar rollout readouts under the same action sequence, while action-relevant state differences remain separable. On a fixed candidate set, bounded ACPC implies bounded candidate-cost drift and preserves the top-ranked candidate within that fixed set under a clean margin condition; ACPC alone admits collapse, motivating a discriminability guard. Across PushT, TwoRoom, Reacher, and Cube, no-noise LeWM is fragile under observation-only Gaussian noise, e.g., PushT falls from 86.33% to 4.33% and Reacher from 58.67% to 18.33%. Full-sequence input-side noise training supplies recovered checkpoints for diagnosis, with bounded replication in PLDM. Paired ACPC-basin probes localize recovery in rollout space (PushT R_F: 1.543 $\to$ 0.088), while rank, transition-resolution, and inverse-dynamics diagnostics check that contraction is not merely collapse. Negative target-view and heteroscedastic-loss ablations expose failure modes. The result is a diagnostic framework and release package, not a new training objective or a closed-loop guarantee.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

1 Introduction

1.1 Latent prediction is not a robustness definition

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [2, 4, 38]. For control, however, "irrelevant" is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose or doorway state if that changes the next useful action. Thus encoder-level invariance is an incomplete target: planning uses the model's *predictions*, so robustness must be stated after action-conditioned prediction.

We use this failure as a controlled diagnostic probe. A no-noise LeWM [26] checkpoint scores 86.33% on unperturbed PushT but only 4.33% under observation-only Gaussian noise at std 0.08 with the goal unperturbed; Reacher drops from 58.67% to 18.33%, and TwoRoom from 94.00% to 65.67%. Observation+goal noise is reported as a harder auxiliary stress condition. Test-time visual fragility is not new [21, 42, 16, 27]; our focus is what it reveals about JEPA latent world-model control.

1.2 Operationalizing action-conditioned predictive consistency

We operationalize a downstream-control view of visual robustness for JEPA world models through *action-conditioned predictive dynamics.* Let $z = E(h)$ and $\tilde{z} = E(\tilde{h})$ be the latents an encoder E produces from an unperturbed history h and a perturbed history $\tilde{h}$ of the same underlying state, and let $F^k(\cdot, \mathbf{a}_{t:t+k-1})$ be the action-conditioned latent predictor rolled out for k steps under an action sequence $\mathbf{a}$. We *allow* $z \neq \tilde{z}$. The robustness target is placed *after* the predictor: in task-relevant coordinates,

$$F^k\big(z, \mathbf{a}_{t:t+k-1}\big) \;\approx\; F^k\big(\tilde{z}, \mathbf{a}_{t:t+k-1}\big), \qquad k = 1, \ldots, H, \tag{1}$$

so that the two branches induce the same next-state and rollout predictions under the same action intervention. The requirement is *selective*: it should hold only for nuisance perturbations of the same underlying state. State differences that change action-conditioned transitions, costs, or the optimal action must remain *discriminable*, or the planner loses the distinctions it needs. We make both halves precise in Section 3.

This diagnostic view changes what counts as a useful robustness measurement. Encoder geometry is a useful first-stage diagnostic because it shows whether visual perturbations enter and reshape the latent neighborhood, but encoder-level latent closeness, $\|z - \tilde{z}\| \to 0$, is not a complete control-robustness definition: large nuisance shifts can wash out after prediction, whereas small shifts can still be amplified downstream. Planning, cost, and action metrics remain useful as *downstream readouts* of predictive consistency; future objectives should regularize action-conditioned predictive dynamics while preserving action-relevant discriminability.

1.3 Existing noisy-evaluation results as a controlled probe

We use input-side noise augmentation as the controlled intervention. The $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ sweep across four tasks creates a reproducible grid of paired baseline/recovered checkpoints for asking where closed-loop recovery appears in the encoder–predictor–planner chain. The downstream correlation analyses condition on $\sigma_{\max}$ to separate sweep-level effects from residual diagnostic signals. Appendix M records a negative target-view ablation: perturbed-history $\to$ original-future denoising is not a sufficient closed-loop fix, so the retained empirical branch is full-sequence perturbed-target training.

1.4 Contributions

C1 — Multi-stage diagnostic principle. We operationalize visual robustness for JEPA world-model control as a multi-stage diagnostic rather than an encoder-invariance score alone. Encoder perturbation geometry measures whether visual noise enters and reshapes the latent state; action-conditioned predictive consistency measures whether that shift changes predicted futures and candidate costs; the discriminability countercondition prevents contraction from erasing action-relevant distinctions.

C2 — Planner link and controlled evidence. Under a Lipschitz cost readout, bounded rollout disagreement bounds candidate-cost drift on a fixed candidate set, and a sufficiently large clean action margin preserves the top-ranked candidate within that fixed set. We use a fixed Gaussian-noise protocol to probe this encoder–predictor–planner chain on LeWM, with PLDM as a second-family replication check.

C3 — Release package and negative checks. We release paired ACPC-basin artifacts, the full LeWM basin grid, cross-checkpoint diagnostic tables, PLDM replication, a target-view ablation, and a heteroscedastic-loss failure case. These artifacts localize what changed at representative high-noise grid rows and expose where stronger readings fail.

2 Related Work

2.1 Latent prediction and JEPA world models

JEPA [23] predicts in latent space rather than reconstructing pixels; I-JEPA [2] and V-JEPA [4, 3] extend this idea to images and video. LeWM [26], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [8]. PLDM [32, 33], via `stable-worldmodel` [25], supplies the second method family in Appendix J. LeJEPA theory [19] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [40, 35, 44]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [38] derive closed-form SSL solutions showing that joint embedding imposes weaker alignment requirements than reconstruction when irrelevant features have large magnitude, while Littwin et al. [24] analyze deep-linear self-distillation dynamics and a

JEPA bias toward high-influence predictive features rather than merely high-variance features. These analyses are representation-level and do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. ACPC is complementary: it asks whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable. Seq-JEPA [12] addresses a related invariance–equivariance tension architecturally. DrQ/DrQ-v2 [21, 42], SODA/DMC-GB [16], DreamerV3 [14], TD-MPC2 [15], ViGMO [27], N-JEPA [28], variational VJEPA [17], and US-JEPA [29] provide visual-robustness and noisy-environment context.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [11], DBC [43], bisimulation-regularized MPC [31], and Bisim-JEPA [37] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [13, 39] and VIBR [7] likewise emphasize downstream control consequences. Wang et al. [41] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [6] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead supplies a JEPA checkpoint diagnostic centered on additive Gaussian pixel-noise stress tests: predictive stability is tested on paired same-state clean/noised rollouts under the same action sequence, fixed-candidate cost stability is derived as a sufficient condition, and a separate discriminability guard checks that action-, transition-, or cost-distinct cases have not been collapsed.

2.4 Representation diagnostics and collapse analysis

We use standard representation diagnostics: effective rank [30, 18, 36], CKA [20], linear probes [1], and anti-collapse context from VICReg [5]. RankMe [10] motivates our cross-checkpoint question: after removing the sweep-level training-noise trend, does a label-free predictor-sensitivity diagnostic retain downstream control signal? The individual metrics are standard; our contribution is the protocol that ties them to closed-loop Gaussian-noise evaluation. A compact boundary table is provided in Appendix A.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability countercondition. Section 3.2 then separates the main descriptive diagnostics from paired ACPC probes.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually perturbed view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation–action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length-k prefix of this sequence after re-indexing time at t, and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \qquad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \tag{2}$$

Let Π denote a *task-relevant predictive readout* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\mathrm{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^{H} \alpha_k \, d\big(\Pi(\hat{z}_{t+k}), \, \Pi(\hat{\tilde{z}}_{t+k})\big), \qquad \alpha_k \geq 0. \tag{3}$$

For the theoretical statements below, we use the induced horizon metric

$$d_H\big((u_1, \dots, u_H), (v_1, \dots, v_H)\big) = \sum_{k=1}^{H} \alpha_k d(u_k, v_k), \tag{4}$$

so that Equation (3) is exactly the clean/perturbed rollout-readout distance under the shared action sequence. In experiments we instantiate this with the L2 distance over rollout tokens used by R_F. In this diagnostic framing, robustness corresponds to small ACPC_H together with action-relevant discriminability, not to the encoder distance $d(z_t, \tilde{z}_t)$ alone. Encoder closeness is a useful first-stage risk signal, but it is neither necessary (a large but task-irrelevant encoder shift can wash out through F_θ and Π) nor sufficient (a small encoder shift can still change the predicted future enough to flip the planned action).

Action-relevant discriminability (countercondition). Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future readout that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future readout diverges,

$$\Delta^H_{\mathrm{dyn}}(s_i, s_j, \mathbf{a}) = d_Y\big(Y^H(s_i, \mathbf{a}), \, Y^H(s_j, \mathbf{a})\big) > m, \tag{5}$$

the representation and predictions should keep them separable,

$$d\big(\Psi(z_i), \, \Psi(z_j)\big) > m', \tag{6}$$

where Ψ is a discriminability readout, possibly action-conditioned (latent, predicted delta under $\mathbf{a}$, inverse-dynamics feature, cost feature, or rollout embedding), and $m, m' > 0$ are margins. Equations (3)–(6) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\mathrm{ACPC}_H \to 0$ by collapsing the latent would violate (6), so the two conditions are inseparable.

3.2 Operational consistency diagnostics

The main paper uses one fixed ACPC instance. For the Gaussian-noise basin diagnostic, Π is the identity readout on the model inference/cost latent rollout, d is Euclidean L2 over rollout tokens, and $H = 8$. The reported quantities are R_E, the median same-state encoder-history spread normalized by the original-view nearest-neighbor L2 scale, and R_F, the median same-state rollout-latent spread normalized by the unperturbed transition L2 reference over the same horizon. R_F/R_E is a descriptive contraction ratio. This empirical R_F is a distributional, recorded-action localization proxy. It should not be read as an estimate of the uniform ϵ in the fixed-candidate stability corollary, which requires a bound over every candidate in a shared set $\mathcal{A}$. The shared-candidate PCC/CRA/MAF readouts in Appendix L are included only as a face-validity check of that downstream link.

Thus R_E and R_F answer different questions. R_E measures whether the visual perturbation changes the encoded neighborhood; R_F measures whether that change survives action-conditioned rollout in coordinates used for planning. Robust checkpoints in this Gaussian-noise study often reduce both, so the claim is not predictor-only contraction. The claim is that encoder contraction must be interpreted through its downstream predictive effect and checked against discriminability.

The main text also keeps three non-ACPC measurements: closed-loop success, the encoder-shift ratio R_E, and proxy discriminability measurements such as effective rank, transition-resolution ratio, and inverse-dynamics probe R^2. Predictive cost consistency, candidate-ranking agreement, margin-conditioned action flips, ADM, and SPRR are exploratory downstream readouts defined in Appendix L.1; they are not used for model selection or for the main robustness claim.

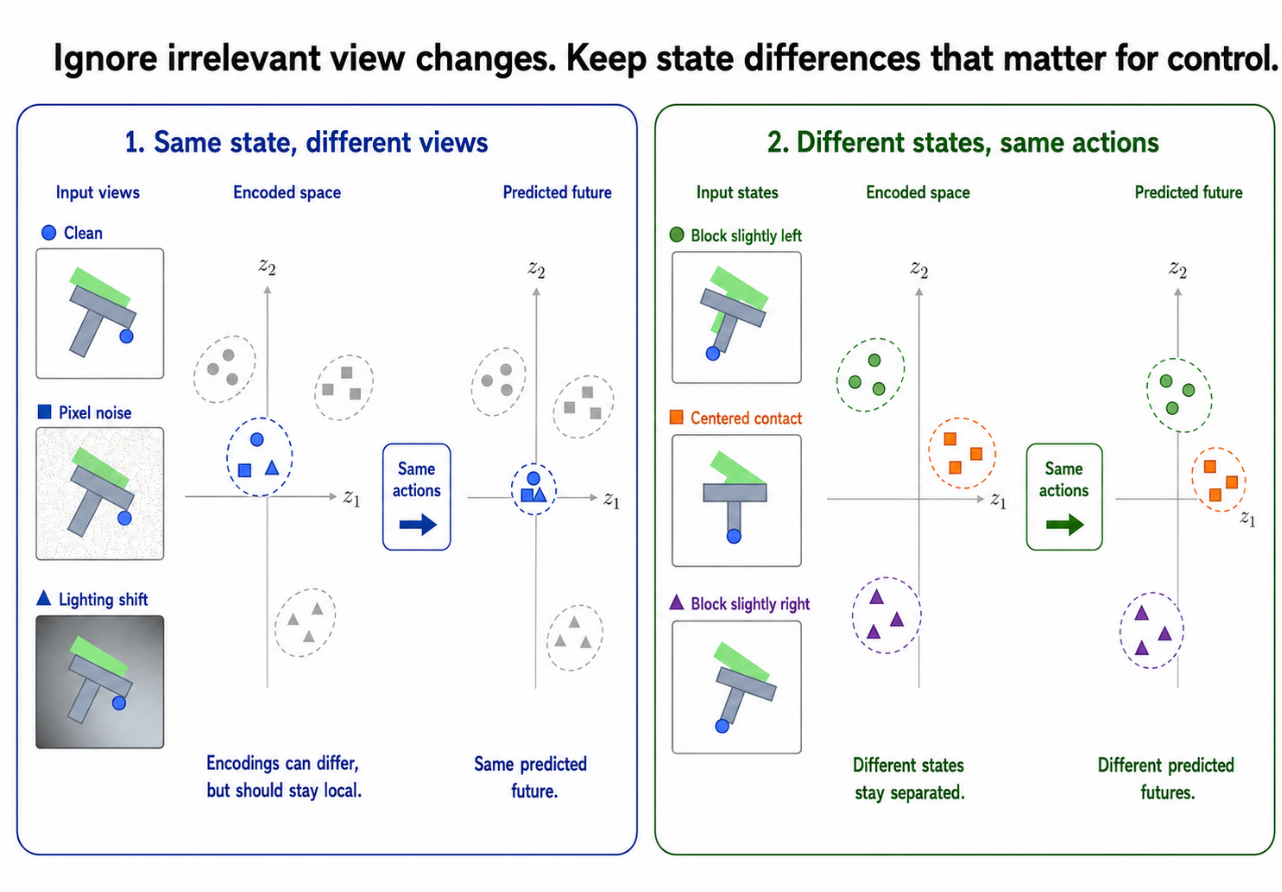


Figure 1: Conceptual reading of the selective-consistency tension. Same-state nuisance variants may encode differently but should induce consistent action-conditioned predictions under the same action sequence, while transition-, cost-, or action-relevant state differences must remain resolvable.

3.3 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding perturbed history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g, define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \ldots, \Pi(\hat{z}_{t+H}), g)\,, \tag{7}$$

where J is the planner's cost readout on the rollout-readout sequence. $C_{\tilde{h}}(\mathbf{a}, g)$ is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Proposition 1 (Encoder shift as one route to rollout disagreement)**.** *Fix an action sequence* $\mathbf{a}$ *and write*

$$G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$$

for the rollout-readout map. If $G_{\mathbf{a}}$ *is locally* L_G*-Lipschitz between* $E_\theta(h)$ *and* $E_\theta(\tilde{h})$ *under encoder metric* d_Z *and rollout-readout metric* d_H*, then*

$$d_H\big(G_{\mathbf{a}}(E_\theta(h)), G_{\mathbf{a}}(E_\theta(\tilde{h}))\big) \leq L_G\, d_Z\big(E_\theta(h), E_\theta(\tilde{h})\big)\,.$$

Proof. This is the local Lipschitz condition applied to the two encoded histories. □

This bound makes encoder geometry a first-stage risk signal: small encoder shifts imply small rollout disagreement only under local insensitivity, large shifts can be contracted by $G_{\mathbf{a}}$, and small shifts can be amplified by the predictor or cost readout. This is why the empirical analysis reports both encoder radii R_E and rollout radii R_F.

Proposition 2 (ACPC controls candidate-cost drift)**.** *Assume that* J *is locally* L_J*-Lipschitz on the clean/perturbed rollout-readout neighborhood evaluated by the fixed candidate set under metric* d_H*. If*

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \leq \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J \epsilon.$$

Proof. The inequality is the local Lipschitz condition applied directly to the clean and perturbed rollout-readout sequences. □

In the reported basin diagnostic, this condition is instantiated with the L2 rollout-token distance underlying R_F; other downstream readouts are treated only as exploratory diagnostics.

Proposition 3 (Candidate top-1 stability)**.** *Let* $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$ *be the shared candidate set. Assume lower cost is preferred, and let* $\mathbf{a}^{(1)}$ *and* $\mathbf{a}^{(2)}$ *denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate* j*,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \geq C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \geq \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. □

Corollary 1 (ACPC-margin condition for candidate stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$ be a shared candidate set and assume the cost readout J is locally L_J-Lipschitz as in Proposition 2. If every candidate in $\mathcal{A}$ has rollout-readout discrepancy at most ϵ between the clean and perturbed branches, and the clean branch has top-1/top-2 margin $\Delta > 2L_J\epsilon$, then the two branches select the same top-1 candidate from $\mathcal{A}$.*

Proof. Proposition 2 gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq L_J\epsilon$ for every candidate. Applying Proposition 3 with $\eta = L_J\epsilon$ gives the result. □

The corollary gives a formal link between ACPC and planning without turning the diagnostic into a closed-loop guarantee. It connects paired predictive agreement to candidate-cost stability and then to action selection under a margin condition. Candidate-ranking agreement and margin-conditioned action flips are downstream readouts of this chain. The result is limited to a fixed candidate set; CEM resampling, repeated replanning, and environment feedback can still change the closed-loop trajectory.

Selective ACPC pseudo-metric. For a nonempty fixed candidate family $\mathcal{A}$, the stability condition can be summarized by

$$D^{\mathcal{A}}_{\text{ACPC}}(h, \tilde{h}) = \max_{\mathbf{a} \in \mathcal{A}} d_H\left(\Pi(F^{1:H}_\theta(E_\theta(h), \mathbf{a})), \Pi(F^{1:H}_\theta(E_\theta(\tilde{h}), \mathbf{a}))\right). \tag{8}$$

When d_H is a metric (or a pseudo-metric, if some horizon weights vanish), $D^{\mathcal{A}}_{\text{ACPC}}$ is a pseudo-metric on histories after passing through the encoder, predictor, and readout: distinct histories can have zero distance if all candidate rollouts are readout-equivalent. This follows because each map $h \mapsto \Pi(F^{1:H}_\theta(E_\theta(h), \mathbf{a}))$ pulls back d_H to a pseudo-metric, and the pointwise maximum of pseudo-metrics is again a pseudo-metric. The useful object is therefore *selective* predictive stability. Small $D^{\mathcal{A}}_{\text{ACPC}}$ for same-state visual perturbations gives candidate-cost and top-1 stability under the preceding Lipschitz and margin assumptions; the same contraction is not desirable for state/action pairs whose external task readouts differ, which is why the discriminability guard is part of the definition rather than an auxiliary aesthetic check.

Corollary 2 (Fixed-candidate predictive stability)**.** *Assume $D^{\mathcal{A}}_{\text{ACPC}}(h, \tilde{h}) \leq \epsilon$ for a same-state visual-perturbation pair and that J is locally L_J-Lipschitz on the rollout readout. If the clean candidate margin satisfies $\Delta > 2L_J\epsilon$, then the clean and perturbed branches select the same top-1 candidate from the fixed set $\mathcal{A}$.*

Proof. By definition of $D^{\mathcal{A}}_{\text{ACPC}}$, every candidate in $\mathcal{A}$ has rollout-readout discrepancy at most ϵ. Proposition 2 bounds the cost drift of every candidate by $L_J\epsilon$, and Proposition 3 gives top-1 stability when $\Delta > 2L_J\epsilon$. □

The statement becomes selective only when it is paired with a discriminability condition such as Equation (6). Without that second condition, fixed-candidate predictive stability can still be achieved by a collapsed encoder–predictor, as shown next.

Encoder closeness is not the right substitute for this condition. It is not necessary: a nuisance-subspace change can move $E_\theta(h)$ and $E_\theta(\tilde{h})$ far apart while leaving $\Pi(F^{1:H}_\theta(\cdot, \mathbf{a}))$ and the cost unchanged. It is not sufficient either: a small encoder displacement can flip a low-margin cost ranking when the predictor or cost readout has high local sensitivity.

Proposition 4 (ACPC alone permits collapse)**.** *There exist encoder–predictor pairs for which* $\text{ACPC}_H = 0$ *for every same-state perturbation pair and every action sequence, while action-distinct states are mapped to identical rollout readouts.*

Proof. Let $E_\theta(h) = z_0$ for every history and let $F^k_\theta(z_0, \mathbf{a}) = u_0$ for every horizon k and action sequence $\mathbf{a}$. Then clean and perturbed branches always have identical rollout readouts, so same-state ACPC is zero. However, any two states that require different actions, costs, or transitions also have identical readouts, violating the discriminability condition in Equation (6). □

Thus low ACPC is meaningful only with a guard on action-relevant separation. This is the role of the transition-resolution and inverse-dynamics probes in the main diagnostics, and it is the failure exposed by the heteroscedastic-loss ablation in Appendix H. This is where ACPC differs from existing nuisance-feature analyses of latent prediction: those results help explain why latent prediction may suppress irrelevant or noisy features at the representation level, whereas the condition above asks whether the remaining perturbation changes action-conditioned rollout readouts and candidate costs while preserving action-relevant separations.

4 Study protocol

4.1 Models and noise intervention

LeWM [26] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization; inference uses CEM for latent-space MPC. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from $\text{Uniform}(0, \sigma_{\max})$. Because the target future frame is also encoded from the noised sequence, this intervention is not an original-target denoising objective. We sweep $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint. Observation+goal Gaussian noise is retained as an auxiliary stress condition.

4.2 Evaluation and diagnostics

The canonical LeWM grid contains 36 epoch-10 checkpoints: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; these are not independent training seeds. Tables report the population mean ± population std across those three evaluation seeds from the canonical evaluation artifact. The runs use a single NVIDIA H800 (80 GB). The main text uses four result blocks: no-noise Gaussian-noise cliffs, the noise-training sweep, paired ACPC-basin radii, and cross-checkpoint diagnostic limitations. The longer five-layer diagnostic definitions, latent-noise mechanism probes, exact implementation keys, and full tables are in Appendix C and Appendix B.10.

For cross-checkpoint checks we compute Spearman ρ across the 9 LeWM checkpoints in each task and partial Spearman conditioned on $\sigma_{\max}$. The partial step removes the sweep-level training-noise trend, asking whether a diagnostic retains residual checkpoint-quality signal. We report 95% percentile bootstrap intervals ($B = 1000$, seed = 42) in the released artifact and use the main table only as a small-sample diagnostic check. PLDM replication is reported in Appendix J.

5 Experiments

The experiments use additive Gaussian pixel noise as the primary controlled stressor for a representation-diagnostic question. We first show that the stressor changes closed-loop behavior. We then ask where recovery appears in the model: encoder geometry (R_E, rank, transition resolution, ID probe), action-conditioned rollout geometry (R_F, rollout drift), and downstream candidate/cost readouts. The goal is not to tune a unique noise level, but to trace how visual perturbations move through the encoder–predictor–planner chain.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep.

5.1 JEPA control fragility under Gaussian observation noise

Table 1 reports LeWM-base success rates under unperturbed and noised evaluation (mean ± population std across three evaluation seeds × 100 evaluations).

Table 1: LeWM-base under test-time Gaussian-noise settings (mean ± population std; 3 evaluation seeds × 100 evaluations). Drop is clean success minus observation-noise 0.08 success.

Task	unpert.	obs 0.05	obs 0.08	obs+goal 0.08	drop
TwoRoom	94.00 ± 3.56	72.33 ± 5.79	65.67 ± 1.25	50.00 ± 1.41	28.33
PushT	86.33 ± 2.36	17.00 ± 1.41	4.33 ± 1.25	4.67 ± 2.05	82.00
Reacher	58.67 ± 1.25	33.67 ± 2.87	18.33 ± 2.05	15.00 ± 2.16	40.33
Cube	66.67 ± 2.62	48.67 ± 6.85	47.00 ± 6.38	46.33 ± 3.68	19.67

LeWM-base is strong on unperturbed images (especially TwoRoom and PushT), but observation noise alone already produces large closed-loop drops: PushT loses 82.00 pts at observation-noise 0.08, Reacher 40.33 pts, TwoRoom 28.33 pts, and Cube 19.67 pts. The observation+goal column shows the stronger full-input-stream

Gaussian-noise stress case; it further degrades TwoRoom and Reacher, but is not the primary robustness endpoint because the goal image is also perturbed. The drop pattern across tasks remains informative: Cube degrades least, while PushT degrades most sharply, consistent with contact-heavy continuous control being most sensitive to visual precision.

PLDM provides a second-family boundary on this failure mode. Without noise augmentation, PLDM drops sharply on PushT (75.33% ± 3.68 unperturbed to 18.33% ± 2.05 at observation-noise 0.08, 57.00 pt) and TwoRoom (96.67% ± 1.70 to 67.00% ± 2.16, 29.67 pt), while Reacher and Cube have much smaller no-noise-training gaps (2.33 and 4.33 pt; Table 17). PLDM Reacher is therefore a useful boundary case: its no-noise baseline is already high, so the LeWM Reacher clean lift below is read as a LeWM fixed-epoch stabilization phenomenon rather than a universal property of noise training. On the PLDM tasks with substantial cliffs, noise-trained rows recover observation-noise performance; Cube remains weakly responsive (Section J). These rows provide replication and boundary evidence for the same diagnostic story.

5.2 Noise augmentation supplies recovered checkpoints for diagnosis

Table 2 summarizes the sweep points needed for the main argument; the complete 8-level tables are in Appendix B.4. All entries use the unified $n = 3$ seeds (42/43/44) × 100 trajectories protocol.

Table 2: LeWM noise-sweep summary. Values are success rate mean ± population std across evaluation seeds. The two point-best columns may refer to different training-noise levels; they summarize the sweep envelope, while the full sweep in Figure 2 and Appendix B.4 provides the grid context.

Task	unpert. base	obs 0.08 base	best clean row	best obs0.08 row	Main reading
TwoRoom	94.00 ± 3.56	65.67 ± 1.25	98.33 ± 0.47 (0.08)	97.67 ± 0.94 (0.08)	large recovery; visually redundant regime
PushT	86.33 ± 2.36	4.33 ± 1.25	89.67 ± 1.70 (0.03)	89.00 ± 4.32 (0.08)	large recovery; contact-sensitive state
Reacher	58.67 ± 1.25	18.33 ± 2.05	86.00 ± 2.94 (0.06)	84.67 ± 5.19 (0.07)	recovery plus fixed-epoch clean lift
Cube	66.67 ± 2.62	47.00 ± 6.38	69.33 ± 0.47 (0.01)	68.33 ± 2.49 (0.03)	weaker recovery signal

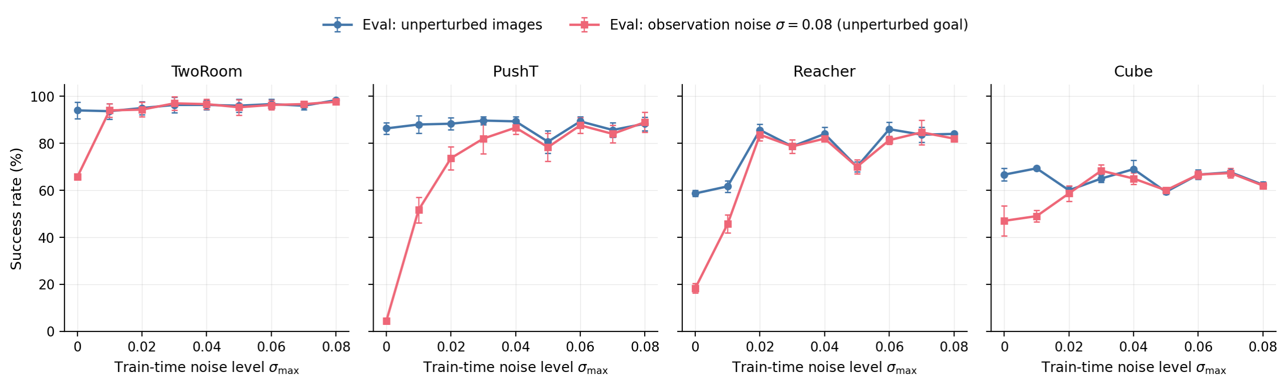


Figure 2: Noise-training sweep across four tasks. Blue circles: unperturbed evaluation; red squares: observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal. Error bars show the population standard deviation across the three evaluation seeds. The curves document the intervention grid and the representative recovered checkpoints used for downstream diagnostics.

The sweep contributes two observations to the main argument. First, input-side noise can be an effective intervention in this protocol: PushT recovers from 4.33% to 89.00% at observation-noise 0.08, and Reacher from 18.33% to 84.67%. Second, the grid supplies representative recovered checkpoints for the downstream diagnostic question: what changed in the encoder–predictor–planner chain?

Reacher also shows a fixed-epoch clean-control lift. Noise training improves not only observation-noise robustness but also unperturbed control: the representative clean row rises from 58.67% at the no-noise baseline to about 86% under noise training. We interpret this as a fixed-protocol regularization or stabilization effect rather than pure robustness recovery. The diagnostic table supports this reading: Reacher does not show a resolution collapse under the fixed $\sigma_{\max} = 0.08$ diagnostic checkpoint, with effective rank and transition resolution flat or slightly improved, while multi-step rollout drift drops sharply ($15.17 \to 0.36$). Thus the same intervention that contracts same-state noisy rollouts coincides with more stable clean action-conditioned prediction at epoch 10. This is a diagnostic observation, not a claim about asymptotic training behavior; independent training seeds and longer learning curves would be needed to separate regularization from optimization timing.

The behavioral evidence for this selective-consistency tension is summarized by Table 2 and fig. 2. Before turning to the broader five-layer diagnostic suite, we report a paired Gaussian-noise ACPC-basin proxy that tests whether same-state noisy views occupy a smaller action-conditioned predictive region after noise training.

5.3 Paired Gaussian-noise ACPC basin diagnostic

We compute an eval-only ACPC basin diagnostic on the same 36 LeWM epoch-10 checkpoints used in the sweep. For each checkpoint we sample 100 fixed dataset windows and create eight paired same-state views using Gaussian pixel noise with std $\in \{0.01, \dots, 0.08\}$, matching the training sweep's noise family. Each original/noised branch is encoded and then rolled out under the same recorded action sequence for horizon $H = 8$. The reported prediction readout is the identity on the resulting inference/cost latent rollout sequence, with L2 distance over rollout tokens. The released source and rendering script are listed in Appendix B.10. The diagnostic intentionally excludes blur/resize so that the ACPC evidence is not confounded by train/eval perturbation-family mismatch.

For each checkpoint, let R_E be the median pairwise spread among the original and noised encoder-history views, normalized by the original-view local NN L2 scale, and let R_F be the median pairwise spread among their $H = 8$ predicted rollout latents, normalized by the unperturbed transition L2 reference over the same horizon. Lower R_F means the same-state perturbation views induce a tighter action-conditioned predictive basin; R_F/R_E is a descriptive contraction ratio rather than a success predictor.

This comparison is a localization diagnostic: the compact rows summarize the high-noise regime visible in closed-loop evaluation, and the basin radius is used afterward to identify where the intervention changed same-state perturbation behavior. Appendix B.5 reports the full grid; Appendix L adds an exploratory shared-candidate downstream check.

Table 3: LeWM Gaussian-noise ACPC-basin proxy: no-noise baseline vs. representative high-noise grid rows. R_E is normalized encoder-view spread and R_F is normalized action-conditioned rollout-view spread under the same action sequence. Drop is clean success minus observation-noise 0.08 success, so negative values indicate no degradation within this evaluation estimate. The rows summarize the high-noise regime; the full grid is reported in Appendix B.5.

Task	checkpoint	obs 0.08	drop	R_E	R_F
TwoRoom	base	65.67	28.33	3.538	0.785
TwoRoom	noise 0.08	97.67	0.67	0.235	0.028
PushT	base	4.33	82.00	1.727	1.543
PushT	noise 0.08	89.00	−0.67	0.137	0.088
Reacher	base	18.33	40.33	4.238	1.012
Reacher	noise 0.07	84.67	−1.00	0.135	0.027
Cube	base	47.00	19.67	1.343	0.957
Cube	noise 0.03	68.33	−3.33	0.238	0.115

The compact rows show the characteristic basin movement in the high-noise regime: R_F is much smaller than at the no-noise baseline on the tasks with large recovery, with PushT decreasing from 1.543 to 0.088 and TwoRoom from 0.785 to 0.028. Figure 3 shows the representative PushT contraction, and Appendix B.8 adds a projection-free local-atlas check. The comparison localizes a joint contraction pattern after the closed-loop sweep: robust checkpoints usually shrink both R_E and R_F, so the result supports rollout-basin localization rather than an encoder-versus-predictor separation.

The basin contraction is not treated as sufficient evidence of robustness. It is paired with the next table's rank, transition-resolution, inverse-dynamics, and heteroscedastic-loss checks to rule out the trivial explanation that the model simply collapsed action-relevant state distinctions. A separate shared-candidate sanity check in Appendix L follows the same direction on the auxiliary observation+goal stress endpoint: for LeWM PushT, ACPC-H/transition drops from 2.740 to 0.149, PCC from 67.3 to 4.6, candidate-ranking agreement rises from 0.323 to 0.986, and margin-conditioned action flips drop from 0.96 to 0.08. Because this check uses the auxiliary observation+goal endpoint and random candidate sets, we keep it as face-validity evidence rather than as the main robustness claim.

The broader representation-level evidence appears next in Table 4.

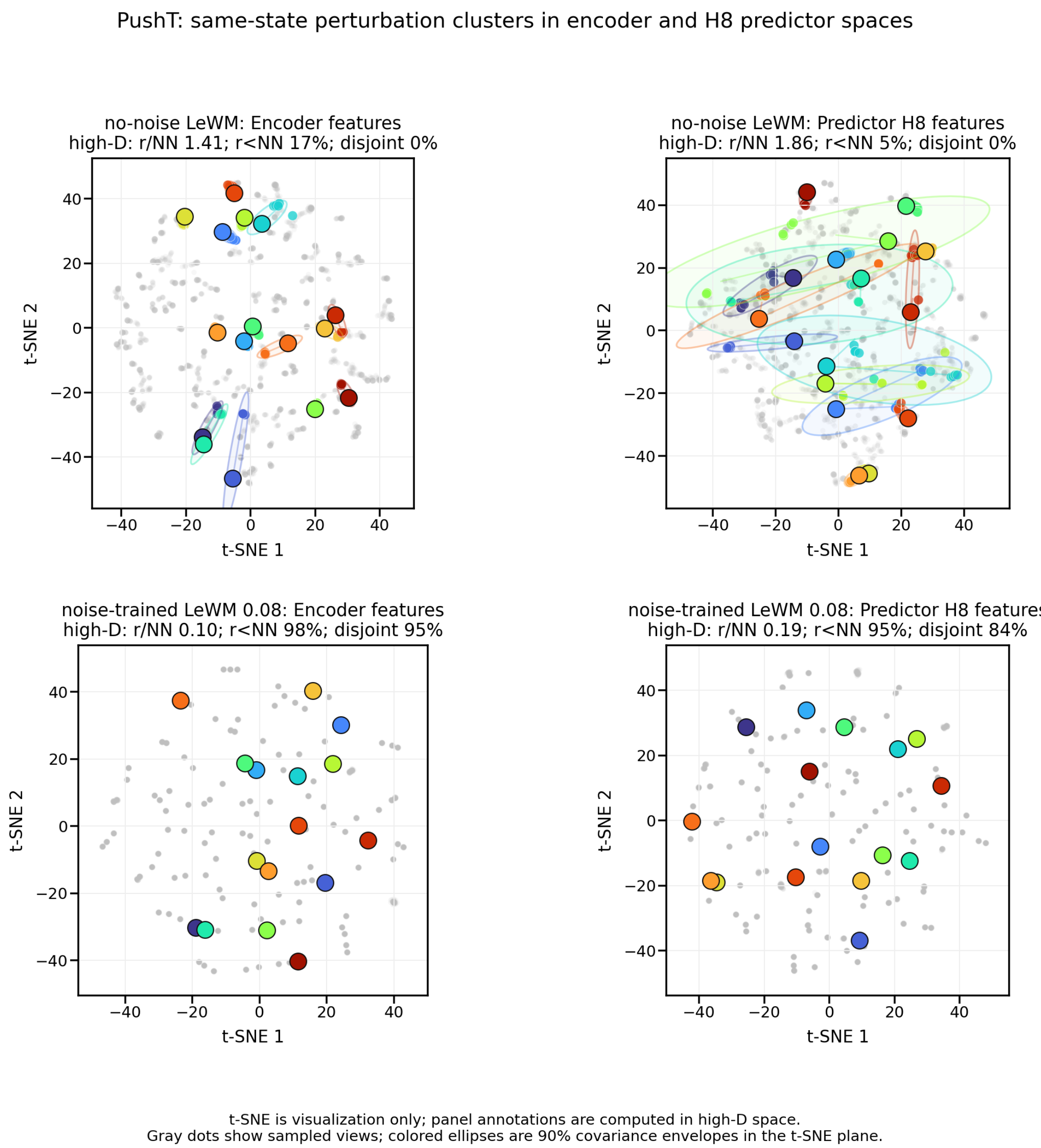


Figure 3: Representative PushT same-state perturbation clusters for Table 3. Top: no-noise-trained LeWM; bottom: noise-trained LeWM at $\sigma_{\max} = 0.08$. Left: encoder-history features; right: $H = 8$ predictor-rollout features. t-SNE coordinates and envelopes are visualization only; the small panel summaries report original-space local statistics.

Table 4: Compact representation-diagnostic movement: LeWM-base → the fixed $\sigma_{\max} = 0.08$ noise-trained checkpoint. The shared row keeps diagnostic comparisons aligned across tasks.

Task	$\sigma_{\max}$	eff. rank	trans. L2	ID probe R^2	rollout T8 L2
TwoRoom	0.08	47.60 → 37.69	0.7216 → 0.6621	0.2889 → 0.1419	18.62 → 0.66
PushT	0.08	76.42 → 78.08	0.3015 → 0.2789	0.7739 → 0.7647	18.65 → 1.06
Reacher	0.08	61.04 → 66.20	0.3704 → 0.3831	0.1621 → 0.1767	15.17 → 0.36
Cube	0.08	73.25 → 74.97	0.4847 → 0.5085	0.6657 → 0.6342	20.20 → 0.60

5.4 Diagnostic analysis: what changes under the intervention

The closed-loop sweep identifies recovered checkpoints under the intervention. The diagnostics ask which latent properties move under noise training and whether basic action-relevant information survives. Table 4 compares core diagnostic metrics on LeWM-base versus the fixed $\sigma_{\max} = 0.08$ noise-trained diagnostic checkpoint for every task. The shared row keeps the cross-task comparison aligned.

Notes. (i) All intervention diagnostics are taken from the corresponding $\sigma_{\max} = 0.08$ checkpoint's per-tool JSON outputs under the diagnostics directory (max-std = 0.1, history-only noise); (ii) the sharp reduction of multi-step rollout drift at this shared high-noise row is not a recording error: the fixed intervention sharply reduces multi-step predictor drift while leaving the single-step task-resolution metrics comparatively flat or mildly improved. This dissociation motivates the cross-checkpoint analysis in Section 5.5, but it is not by itself evidence of better controllability.

Mechanistic reading.

- **TwoRoom.** Low-dimensional, discrete, visually redundant. The observed compression (effective rank 47.6 → 37.7) is compatible with high success in this task; the data do not show that compactness itself causes the improvement.
- **PushT.** Continuous contact requires fine-grained pose resolution. Noise training largely respects this bound even at the fixed $\sigma_{\max} = 0.08$ diagnostic checkpoint: effective rank and inverse-dynamics probe stay flat (76.4 → 78.1; 0.774 → 0.765), while the L2 transition-resolution metric declines mildly (0.3015 → 0.2789). The catastrophic resolution erasure on PushT appears under error-based reweighting (Section H), not under input-side noise.
- **Reacher.** Low-dimensional continuous reaching does not show a resolution collapse in Table 4: effective rank, transition resolution, and the controllability probe remain flat or slightly improve. The notable change is the large reduction in multi-step predictor drift (15.17 → 0.36), matching the later finding that Reacher's residual noise-drop signal lives in the rollout metric rather than in the single-step fragility ratio. This also explains why Reacher's unperturbed score rises under the fixed epoch-10 protocol: the noise-trained checkpoint coincides with more stable rollout prediction without erasing the low-dimensional control state. Because this is one trained checkpoint per noise level, we do not interpret the clean lift as an asymptotic training advantage.
- **Cube.** Cube is the guardrail against over-reading rollout drift. The fixed $\sigma_{\max} = 0.08$ checkpoint sharply reduces multi-step predictor drift (20.20 → 0.60) while rank and transition resolution are flat or slightly higher (73.3 → 75.0; 0.4847 → 0.5085). Its closed-loop recovery remains weaker than the large-cliff tasks, so predictor stabilization alone is not sufficient evidence of robust control.
- **Predictor-rollout caveat.** A drop in multi-step predictor rollout drift is not unambiguously good news. It can also mean the latent has become more *predictable* without being more *controllable*: predictor stability bought by sacrificing resolution.

These proxy discriminability checks are the main-text guard against a collapse-only reading. PushT is the sharpest case: the fixed high-noise input-side checkpoint keeps effective rank and ID probe essentially flat, while the heteroscedastic-loss ablation in Appendix H shows the failure mode when error-based reweighting suppresses action-critical contact transitions. We therefore claim a proxy-level discriminability guard, not an oracle margin proof.

5.5 Cross-checkpoint correlation analysis

We analyze two single-value-per-checkpoint diagnostic metrics that have full coverage on all 9 checkpoints per task: the fragility ratio, defined in Section C.2 as the single-step predictor target shift normalized by the nearest-neighbor cosine distance at the diagnostic's maximum injected std, and the corresponding multi-step rollout L2

Table 5: Partial Spearman ρ(metric, noise drop | $\sigma_{\max}$), $n = 9$ per task.

Metric	TwoRoom	PushT	Reacher	Cube
fragility ratio	-0.03	$+0.19$	$+0.27$	$+0.12$
rollout drift	0.00	-0.01	$+0.37$	$+0.23$

drift. Both are pure checkpoint-level quantities derived from predictor-sensitivity diagnostics, independent of the evaluation protocol; exact artifact names are listed in Appendix B.10.

Unconditioned Pearson/Spearman sweep-trend correlations are reported in Appendix E for transparency, but they are dominated by the monotone training-noise sweep. The main residual check is therefore partial Spearman conditioned on $\sigma_{\max}$:

After removing the monotone sweep trend associated with $\sigma_{\max}$, the **fragility metric carries little stable information about the size of the noise drop** on any task; the bootstrap intervals are wide and include, or round to, zero. The **multi-step predictor drift** retains only weak residual signal under the unperturbed-goal observation-noise 0.08 endpoint (Reacher +0.37, Cube +0.23, both with wide intervals). This weakens the older observation+goal stress reading and is the reason we treat the cross-checkpoint diagnostic as a local explanation aid, not as a robustness predictor.

The PushT detailed table and scatter plot are moved to Appendix F. They show the same split at higher resolution: fragility ratio carries residual checkpoint-quality signal for unperturbed and noisy success after conditioning on $\sigma_{\max}$, but it does not isolate the Gaussian-noise gap itself.

Appendix I reports the latent-noise mechanism probe. Its role is supporting, not central: the strongest common signal is encoder-side perturbation transduced by the predictor, but the present data do not isolate planner or cost-surface causality.

6 Discussion and limitations

Scope. This paper is a controlled diagnostic study. The canonical LeWM grid uses three evaluation seeds per checkpoint, not independent training seeds. Gaussian pixel noise is the main training and evaluation axis; blur is an eval-only sanity check in Appendix K. The ACPC basin probe is computed after closed-loop evaluation identifies the reported high-noise grid rows and serves as an after-the-fact localization diagnostic. The discriminability condition is tested through proxy diagnostics rather than oracle state margins. The main mechanism analysis is LeWM-centered, with PLDM used as a replication and scope check.

Interpretation. The data challenge a too-strong reading of latent prediction: predicting future latents does not by itself guarantee robustness to visual noise. Encoder invariance is useful but incomplete. The useful target is agreement of action-conditioned predictions in task-relevant coordinates while retaining distinctions that affect actions, costs, or transitions. This reading is compatible with the task variation: TwoRoom can tolerate substantial nuisance contraction, PushT contact control needs fine pose resolution, Reacher sits in a lower-dimensional continuous regime, and Cube is less sensitive to global Gaussian noise in this sweep.

Using the diagnostics. The diagnostics can help triage checkpoints within this protocol, but final model choice still requires closed-loop Gaussian-noise evaluation. The PushT partial-correlation split is the main warning: after conditioning on $\sigma_{\max}$, fragility ratio still carries residual checkpoint-quality signal for unperturbed and noisy success, yet it does not isolate the Gaussian-noise gap. The ACPC basin table has the same status. The fixed-0.08 diagnostic table localizes what changed under a shared high-noise intervention, while the full grid in Appendix B.5 shows that the basin radius is not a standalone success predictor.

Negative checks and next steps. Two negative checks bound the method story. The heteroscedastic NLL probe collapses PushT because contact-control transitions are high-error and action-critical; prediction difficulty alone is the wrong gate. The target-view ablation in Appendix M rules out perturbed-history $\rightarrow$ original-future targets as a sufficient closed-loop fix. The natural next experiment is therefore a trained paired predictive-dynamics consistency objective after the action-conditioned predictor, gated by action sensitivity and protected by a discriminability guard. Broader replication should include frozen or pretrained visual features, EMA-target or variational JEPA variants, and non-Gaussian training perturbations.

From diagnostics to more generalizable world models. The Gaussian-noise study should be read as a diagnostic probe, not as an endpoint. It points to two foundation capabilities for more generalizable latent world models. First, the encoder should filter visual nuisance variation while preserving geometry, contact, goal relations, and other action-relevant state differences needed for planning. Second, the predictor should make these representations useful under intervention: clean and perturbed views of the same state should produce consistent action-conditioned rollouts and cost-relevant predictions under the same action sequence, while states or actions that imply different futures remain separable.

These requirements are broader than additive Gaussian noise. They are prerequisites for world models that can adapt predictions and plans under new appearances, goals, or dynamics from context rather than memorizing a fixed training distribution. This paper does not test such context adaptation directly; it provides diagnostics that expose where the current encoder–predictor–planner chain succeeds, where it collapses nuisance variation safely, and where stronger objectives must protect action-relevant distinctions.

7 Conclusion

This paper studies Gaussian-noise robustness in JEPA latent world-model control as a multi-stage diagnostic problem. Encoder geometry shows how visual perturbations enter the latent state, action-conditioned rollout consistency shows whether they affect predicted futures and candidate costs, and discriminability checks whether robustness is bought by erasing action-relevant distinctions. Controlled additive-noise tests show that latent prediction alone does not guarantee closed-loop robustness. Input-side noise augmentation supplies recovered checkpoints for analyzing this failure mode. The paired ACPC basin diagnostic adds a local diagnostic readout at representative high-noise grid rows: same-action prediction radii are smaller under same-state perturbations, while pointwise fragility does not isolate the Gaussian-noise gap after conditioning on train-time noise.

The target-view ablation in Appendix M rules out perturbed-history $\rightarrow$ original-future targets as a sufficient closed-loop fix and leaves full-sequence perturbed targets as the empirical branch for this study. The broader implication is diagnostic rather than solved generalization: more generalizable world models need control-relevant representations and action-conditioned predictive dynamics that stay stable under nuisance variation without collapsing distinctions needed for planning. The next method question is paired predictive-dynamics consistency after the action-conditioned predictor, gated by action sensitivity and protected by discriminability.

Acknowledgements

Code, aggregate evaluation artifacts, rendering scripts, and data/checkpoint pointers for this revision are available at `https://github.com/Anguo-star/le-wm`; JSON artifact hashes are listed in the data manifest. We thank the LeWM authors for releasing the baseline framework on which this study builds.

References

[1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. ICLR 2017 Workshop Track / arXiv preprint, 2017.

[2] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023.

[3] Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zholus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025.

[4] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mido Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *Transactions on Machine Learning Research*, 2024.

[5] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.

[6] Tianxiang Cheng, Shiqi Deng, Jialong Li, Shengjie Wang, Jiange Yang, Xuanqi He, Baochun Li, and Anqi Li. Reimagination with observation intervention for robust visual model predictive control, 2025.

[7] Tom Dupuis, Jaonary Rabarisoa, Quoc-Cuong Pham, and David Filliat. VIBR: Learning view-invariant value functions for robust visual control. In *Proceedings of The 2nd Conference on Lifelong Learning Agents*, volume 232 of *Proceedings of Machine Learning Research*, pages 658–682. PMLR, 2023.

[8] T. W. Epps and Lawrence B. Pulley. A test for normality based on the empirical characteristic function. *Biometrika*, 70(3):723–726, 1983.

[9] Carlos E. García, David M. Prett, and Manfred Morari. Model predictive control: Theory and practice — a survey. *Automatica*, 25(3):335–348, 1989.

[10] Quentin Garrido, Randall Balestriero, Laurent Najman, and Yann LeCun. RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 10929–10974. PMLR, 2023.

[11] Carles Gelada, Saurabh Kumar, Jacob Buckman, Ofir Nachum, and Marc G. Bellemare. DeepMDP: Learning continuous latent space models for representation learning. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2170–2179. PMLR, 2019.

[12] Hafez Ghaemi, Eilif Benjamin Muller, and Shahab Bakhtiari. seq-jepa: Autoregressive predictive learning of invariant-equivariant world models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[13] Christopher Grimm, Andre Barreto, Satinder Singh, and David Silver. The value equivalence principle for model-based reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.

[14] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640(8059):647–653, 2025.

[15] Nicklas Hansen, Hao Su, and Xiaolong Wang. Td-mpc2: Scalable, robust world models for continuous control. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

[16] Nicklas Hansen and Xiaolong Wang. Generalization in reinforcement learning by soft data augmentation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 13611–13617, 2021.

[17] Yongchao Huang. Vjepa: Variational joint embedding predictive architectures as probabilistic world models. *arXiv preprint arXiv:2601.14354*, 2026.

[18] Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.

[19] David Klindt, Yann LeCun, and Randall Balestriero. When does lejepa learn a world model?, 2026.

[20] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3519–3529. PMLR, 2019.

[21] Ilya Kostrikov, Denis Yarats, and Rob Fergus. Image augmentation is all you need: Regularizing deep reinforcement learning from pixels. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.

[22] Dirk P. Kroese, Sergey Porotsky, and Reuven Y. Rubinstein. The cross-entropy method for continuous multi-extremal optimization. *Methodology and Computing in Applied Probability*, 8(3):383–407, 2006.

[23] Yann LeCun. A path towards autonomous machine intelligence. OpenReview preprint, 2022.

[24] Etai Littwin, Omid Saremi, Madhu Advani, Vimal Thilak, Preetum Nakkiran, Chen Huang, and Joshua Susskind. How JEPA avoids noisy features: The implicit bias of deep linear self distillation networks, 2024.

[25] Lucas Maes, Quentin Le Lidec, Luiz Facury, Nassim Massaudi, Ayush Chaurasia, Francesco Capuano, Richard Gao, Taj Gillin, Dan Haramati, Damien Scieur, Yann LeCun, and Randall Balestriero. stable-worldmodel: A platform for reproducible world modeling research and evaluation, 2026.

[26] Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. Leworldmodel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026.

[27] Mingyu Park, Samyeul Noh, Hyun Myung, and Donghwan Lee. Zero-shot visual generalization in model-based reinforcement learning via latent consistency. OpenReview (ICLR 2026 submission), 2025.

[28] Yuping Qiu, Rui Zhu, and Ying-cong Chen. Improving joint embedding predictive architecture with diffusion noise. *arXiv preprint arXiv:2507.15216*, 2025.

[29] Ashwath Radhachandran, Vedrana Ivezić, Shreeram Athreya, Ronit Anilkumar, Corey W. Arnold, and William Speier. Us-jepa: A joint embedding predictive architecture for medical ultrasound. *arXiv preprint arXiv:2602.19322*, 2026.

[30] Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *Proceedings of the 15th European Signal Processing Conference (EUSIPCO)*, pages 606–610, 2007.

[31] Yutaka Shimizu and Masayoshi Tomizuka. Bisimulation metric for model predictive control. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.

[32] Vlad Sobal, Jyothir S V, Siddhartha Jalagam, Nicolas Carion, Kyunghyun Cho, and Yann LeCun. Joint embedding predictive architectures focus on slow features, 2022.

[33] Vlad Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestriero, Tim G. J. Rudner, and Yann LeCun. Stress-testing offline reward-free reinforcement learning: A case for planning with latent dynamics models. In *WRL@ICLR 2025*, 2025.

[34] Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *Proceedings of the International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, pages 20827–20840. PMLR, 2022.

[35] Alex Tamkin, Margalit Glasgow, Xiluo He, and Noah Goodman. Feature dropout: Revisiting the role of augmentations in contrastive learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[36] Jayden Teoh, Manan Tomar, Kwangjun Ahn, Edward S. Hu, Pratyusha Sharma, Akshay Krishnamurthy, Riashat Islam, Alex Lamb, and John Langford. Next-latent prediction transformers learn compact world models. *arXiv preprint arXiv:2511.05963*, 2025.

[37] Leonardo F. Toso, Davit Shadunts, Yunyang Lu, Nihal Sharma, Donglin Zhan, Nam H. Nguyen, and James Anderson. Learning invariant visual representations for planning with joint-embedding predictive world models. *arXiv preprint arXiv:2602.18639*, 2026.

[38] Hugues Van Assel, Mark Ibrahim, Tommaso Biancalani, Aviv Regev, and Randall Balestriero. Joint embedding vs reconstruction: Provable benefits of latent space prediction for self-supervised learning, 2025.

[39] Claas A. Voelcker, Anastasiia Pedan, Arash Ahmadian, Romina Abachi, Igor Gilitschenski, and Amir-Massoud Farahmand. Calibrated value-aware model learning with probabilistic environment models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 61745–61768. PMLR, 2025.

[40] Tongzhou Wang and Phillip Isola. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 9929–9939. PMLR, 2020.

[41] Zijie Wang, Wei Zhang, Weiming Zhang, Fanqi Zhang, Xiao Tan, Yipeng Qin, and Guanbin Li. World models as group actions, 2026.

[42] Denis Yarats, Rob Fergus, Alessandro Lazaric, and Lerrel Pinto. Mastering visual continuous control: Improved data-augmented reinforcement learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.

[43] Amy Zhang, Rowan T. McAllister, Roberto Calandra, Yarin Gal, and Sergey Levine. Learning invariant representations for reinforcement learning without reconstruction. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.

[44] Junbo Zhang and Kaisheng Ma. Rethinking the augmentation module in contrastive learning: Learning hierarchical augmentation invariance with expanded views. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16650–16659, 2022.

A Related-work boundary table

Table 6: Boundary to neighboring consistency and control-representation work.

Neighboring thread	Shared concern	Boundary in this paper
ViGMO [27]	Visual distractions and latent consistency	We do not propose a generic encoder-consistency regularizer; our diagnostic places consistency after action-conditioned rollout and adds a discriminability guard.
ReOI [6]	Robust visual MPC under distractors	We do not propose an observation-intervention policy; we audit existing JEPA checkpoints with paired same-action clean/noised rollouts and fixed-candidate stability conditions.
Bisimulation / Bisim-JEPA [11, 43, 37]	Control-relevant state abstraction	Same-state perturbed and unperturbed views need not share an encoder latent; agreement is required at the predictive-dynamics level.
LeJEPA theory [19]	Latent variables and latent planning	We do not claim identifiability; we test whether learned checkpoints preserve predictive consistency under visual perturbation.
Group-action world models [41]	Action-faithful rollout structure	We test same-action paired perturbation consistency, not identity/inverse/composition consistency across action sequences.
Value-aware models [13, 39]	Models should serve downstream control	We make this operational as paired action-conditioned rollout agreement, not value equivalence or value prediction.
V-JEPA family [4, 3, 17]	Latent video prediction and noisy-environment probes	Representation recovery and closed-loop control stress different properties; our probe measures eval-time closed-loop behavior.

B Experimental details

B.1 Environments

- **PushT.** 2D continuous pushing; 20,000 expert episodes; $\sim$196 steps/episode; action dim 2 (direction + magnitude); 224×224 RGB.

- **TwoRoom.** 2D continuous navigation; 10,000 episodes; ~ 92 steps; action dim 2; 224×224 RGB.
- **Reacher.** 2D arm control (DeepMind Control Suite); 10,000 episodes; 200 steps; action dim 2.
- **Cube.** 3D cube manipulation (OGBench); 10,000 episodes; 200 steps; action dim 7.

B.2 Noise augmentation implementation

The actual implementation is `utils.py::AddNormalizedGaussianNoise`. Key points: (i) noise is added on ImageNet-normalized tensors and must be divided by the per-channel std to retain "pixel-space std" semantics; (ii) sampling is per-frame independent over the leading frame dims, not per-batch; (iii) all sweeps in this paper fix the noise probability to 1.0, set the lower std to 0, and vary only the upper std $\sigma_{\max}$.

```
class AddNormalizedGaussianNoise:
    """Per-frame independent. Each frame draws Bernoulli(noise_prob) then
    std ~ Uniform(std_min, std_max). Pixel-space std is converted to
    normalized space by dividing by per-channel std before adding."""
    def __init__(self, std_min, std_max, noise_prob=1.0):
        self.std_low, self.std_high, self.noise_prob = std_min, std_max, noise_prob
        self.channel_std = torch.as_tensor(IMAGENET_STD) # (C,)

    def __call__(self, x):
        if self.std_high <= 0 or self.noise_prob <= 0:
            return x
        leading = x.shape[:-3]
        stds = torch.empty(leading, device=x.device, dtype=x.dtype).uniform_(
            self.std_low, self.std_high)
        if self.noise_prob < 1.0:
            mask = (torch.rand(leading, device=x.device) < self.noise_prob).to(x.dtype)
            stds = stds * mask
        per_frame_scale = stds.view(*leading, 1, 1, 1)
        channel_factor = (1.0 / self.channel_std.to(x.device, x.dtype)).view(
            *([1] * len(leading)), -1, 1, 1)
        scale = per_frame_scale * channel_factor # pixel-space std -> normalized
        return x + torch.randn_like(x) * scale
```

B.3 Evaluation protocol

Unperturbed evaluation uses no noise, 100 trajectories per seed $\times$ 3 seeds (42/43/44), totalling 300 trajectories per condition. The primary noisy evaluation adds Gaussian noise of std $\in \{0.05, 0.08\}$ to observation pixels while leaving the goal image unperturbed under the same 3-seed protocol. Observation+goal Gaussian noise is retained as an auxiliary full-input-stream stress condition and is reported separately where used. Success criterion is task-specific (PushT: T-block pose match; TwoRoom: target region; Reacher: joint-angle match; Cube: cube position match).

B.4 Full LeWM noise-sweep tables

Tables 7 and 8 give the complete LeWM sweep behind Table 2 and fig. 2. All cells are success rate mean $\pm$ population std across the three evaluation seeds.

Table 7: LeWM+noise sweep, unperturbed success rate (%).

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	94.00 ± 3.56	86.33 ± 2.36	58.67 ± 1.25	66.67 ± 2.62
0.01	93.67 ± 3.30	88.00 ± 3.74	61.67 ± 2.49	69.33 ± 0.47
0.02	95.00 ± 2.83	88.33 ± 2.62	85.67 ± 2.49	60.00 ± 1.63
0.03	96.33 ± 3.30	89.67 ± 1.70	78.67 ± 1.25	65.00 ± 1.63
0.04	96.33 ± 2.05	89.33 ± 2.05	84.00 ± 2.94	69.00 ± 3.74
0.05	96.00 ± 2.83	80.67 ± 4.78	70.00 ± 2.16	59.33 ± 0.94
0.06	96.67 ± 2.05	89.33 ± 2.05	86.00 ± 2.94	66.67 ± 2.05
0.07	96.00 ± 1.63	85.67 ± 3.09	83.67 ± 3.30	67.67 ± 0.94
0.08	98.33 ± 0.47	88.33 ± 2.87	84.00 ± 0.82	62.33 ± 1.25

Table 8: LeWM+noise sweep, observation-noise 0.08 success rate with unperturbed goal (%).

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	65.67 ± 1.25	4.33 ± 1.25	18.33 ± 2.05	47.00 ± 6.38
0.01	94.00 ± 2.83	51.67 ± 5.44	45.67 ± 3.86	49.00 ± 2.45
0.02	94.33 ± 3.09	73.67 ± 4.99	83.67 ± 2.49	58.67 ± 3.30
0.03	97.00 ± 2.94	82.00 ± 6.48	78.67 ± 2.87	68.33 ± 2.49
0.04	96.67 ± 2.05	86.67 ± 2.87	82.00 ± 0.82	65.00 ± 2.45
0.05	95.33 ± 3.30	78.33 ± 5.91	70.00 ± 2.94	60.00 ± 0.82
0.06	96.33 ± 2.05	87.67 ± 3.30	81.33 ± 1.70	66.67 ± 1.70
0.07	96.67 ± 1.25	84.00 ± 3.74	84.67 ± 5.19	67.33 ± 2.05
0.08	97.67 ± 0.94	89.00 ± 4.32	82.00 ± 1.41	62.00 ± 1.41

B.5 Full LeWM ACPC-basin grid

Table 9: Full LeWM Gaussian-noise ACPC-basin grid. Success is observation-noise 0.08 success with an unperturbed goal; drop is unperturbed success minus that success. Negative drop means no degradation within this evaluation estimate. R_E and R_F are the normalized encoder-history and $H = 8$ rollout-view radii used in Table 3.

Task	$\sigma_{\max}$	unpert.	obs 0.08	drop	R_E	R_F	R_F/R_E	note
TwoRoom	0	94.00	65.67	28.33	3.538	0.785	0.222	base
TwoRoom	0.01	93.67	94.00	-0.33	1.633	0.232	0.142	
TwoRoom	0.02	95.00	94.33	0.67	0.931	0.124	0.133	
TwoRoom	0.03	96.33	97.00	-0.67	0.720	0.084	0.117	
TwoRoom	0.04	96.33	96.67	-0.33	0.518	0.056	0.108	
TwoRoom	0.05	96.00	95.33	0.67	0.360	0.039	0.108	
TwoRoom	0.06	96.67	96.33	0.33	0.300	0.034	0.112	
TwoRoom	0.07	96.00	96.67	-0.67	0.259	0.029	0.112	
TwoRoom	0.08	98.33	97.67	0.67	0.235	0.028	0.121	obs0.08 ref
PushT	0	86.33	4.33	82.00	1.727	1.543	0.893	base
PushT	0.01	88.00	51.67	36.33	0.677	0.474	0.701	
PushT	0.02	88.33	73.67	14.67	0.383	0.250	0.654	
PushT	0.03	89.67	82.00	7.67	0.253	0.177	0.699	
PushT	0.04	89.33	86.67	2.67	0.222	0.145	0.653	
PushT	0.05	80.67	78.33	2.33	0.232	0.141	0.607	
PushT	0.06	89.33	87.67	1.67	0.169	0.106	0.628	
PushT	0.07	85.67	84.00	1.67	0.159	0.105	0.663	
PushT	0.08	88.33	89.00	-0.67	0.137	0.088	0.642	obs0.08 ref
Reacher	0	58.67	18.33	40.33	4.238	1.012	0.239	base
Reacher	0.01	61.67	45.67	16.00	1.359	0.241	0.177	
Reacher	0.02	85.67	83.67	2.00	0.067	0.017	0.251	
Reacher	0.03	78.67	78.67	0.00	0.283	0.052	0.185	
Reacher	0.04	84.00	82.00	2.00	0.200	0.038	0.191	
Reacher	0.05	70.00	70.00	0.00	0.064	0.015	0.241	
Reacher	0.06	86.00	81.33	4.67	0.147	0.031	0.209	
Reacher	0.07	83.67	84.67	-1.00	0.135	0.027	0.200	obs0.08 ref
Reacher	0.08	84.00	82.00	2.00	0.116	0.024	0.209	
Cube	0	66.67	47.00	19.67	1.343	0.957	0.713	base
Cube	0.01	69.33	49.00	20.33	0.772	0.471	0.610	
Cube	0.02	60.00	58.67	1.33	0.063	0.019	0.293	
Cube	0.03	65.00	68.33	-3.33	0.238	0.115	0.481	obs0.08 ref
Cube	0.04	69.00	65.00	4.00	0.177	0.078	0.440	
Cube	0.05	59.33	60.00	-0.67	0.045	0.010	0.232	
Cube	0.06	66.67	66.67	0.00	0.120	0.046	0.385	
Cube	0.07	67.67	67.33	0.33	0.098	0.038	0.390	
Cube	0.08	62.33	62.00	0.33	0.094	0.036	0.385	

B.6 Compute

Training runs on a single NVIDIA H800 (80 GB) and follows the LeWM training schedule released with the baseline.

B.7 Main-figure rendering

Figures 2 and 6 are produced by the main plot script listed in Appendix B.10; no checkpoint or model loading is required.

Figure 3 is produced by `tools/paper1_selective_contraction.py`. The rendering command used for this paper is:

```
python -m tools.paper1_selective_contraction \
  --plot-clusters --plot-tasks PushT \
  --n-sequences 128 --cluster-anchor-count 16 \
  --view-stds 0.0 0.01 0.04 0.08 \
  --cluster-perturb-repeats 6 \
  --cluster-out-dir assets/paper1_figs \
  --cluster-envelope ellipse --cluster-envelope-coverage 0.90
```

Unlike the aggregate figure generator, this optional rendering path loads the PushT checkpoints and dataset windows to draw repeated same-state perturbation samples. The small panel summaries report high-dimensional statistics; the t-SNE ellipses are only qualitative 2-D covariance envelopes.

B.8 Local cluster atlas: a projection-free check of the basin visualization

Because t-SNE distorts global distances, Figure 4 provides a complementary *local normalized atlas* view of the same PushT data. Each small box selects one original dataset state; the circle radius equals the distance to its nearest *other* original state, computed in the original high-dimensional feature space, and the color markers are that state's original and Gaussian-perturbed views after a local PCA used only for in-box display. The visual question per box is therefore exactly the high-dimensional one: does the same-state perturbation cloud stay inside the nearest-different-state radius? On the no-noise-trained checkpoint many perturbation views escape their circles (median $r/\mathrm{NN} > 1$, few disjoint balls); on the full-sequence noise-trained checkpoint the clouds contract well inside the circles in both encoder and predictor-rollout features. This avoids the global-projection caveats of Figure 3 at the cost of showing only local neighborhoods; the quantitative multi-task evidence remains Table 3.

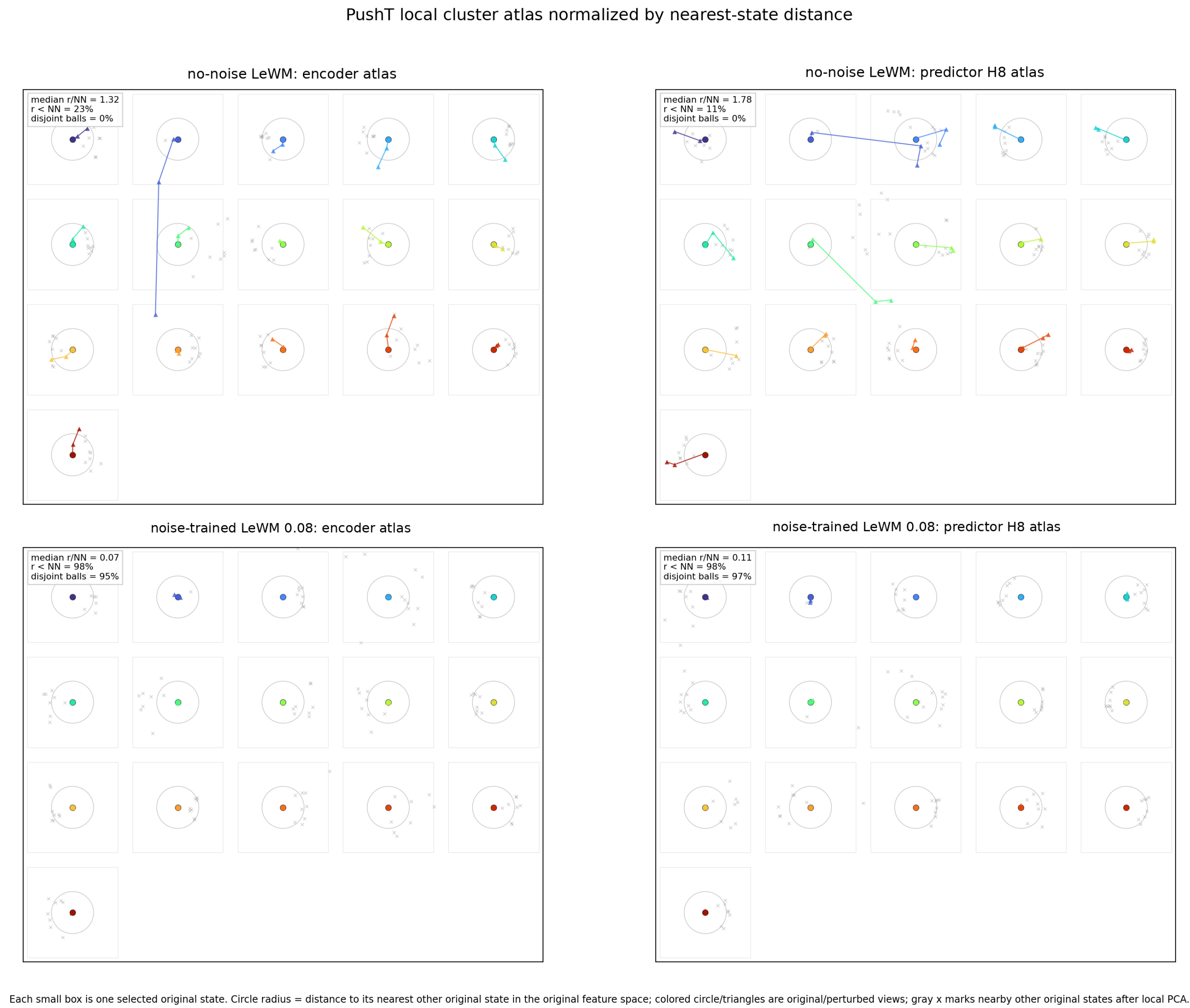


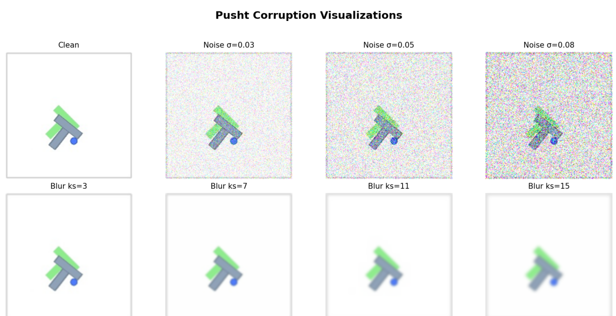
Figure 4: PushT local cluster atlas ($n = 128$ windows, 16 anchors, 8 neighbors). Each box is one original state with Gaussian-perturbed views after local PCA; the circle radius is the high-dimensional nearest-other-state distance. Panel summaries report original-space local statistics.

The atlas rendering command is:

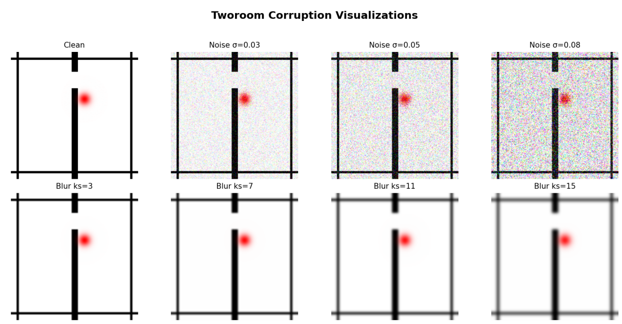
```
python -m tools.paper1_selective_contraction \
  --plot-atlas --plot-tasks PushT \
  --n-sequences 128 --atlas-anchor-count 16 \
  --view-stds 0.0 0.01 0.04 0.08 \
  --atlas-out-dir assets/paper1_figs
```

B.9 Perturbation visualizations

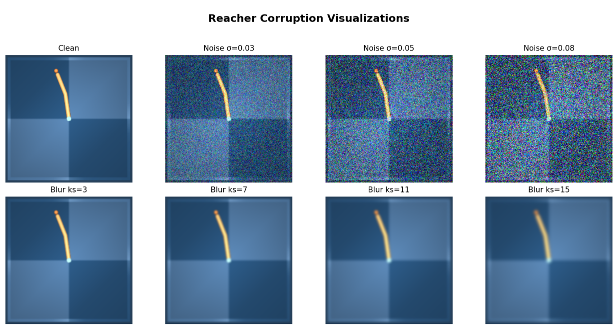
Figure 5 shows representative renderings of the two evaluated perturbation families, applied to a sample observation from each of the four tasks. The Gaussian-noise row includes a mild calibration severity $\sigma = 0.03$ in addition to the two evaluation endpoints $\sigma \in \{0.05, 0.08\}$ used in Sections 5.1 and 5.2; the main sweep treats observation-only noise with an unperturbed goal as the primary endpoint and keeps observation+goal noise as an auxiliary stress condition. The Gaussian-blur kernel sizes ($k \in \{3, 7, 11, 15\}$) match the stress test in Section K. Each panel is a single 224×224 RGB frame from the corresponding task's evaluation rollout, after perturbation and before encoder normalization.



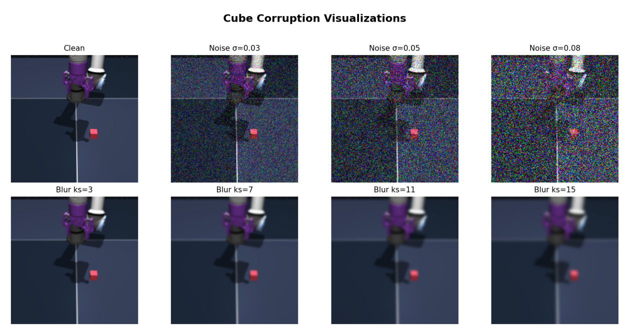
(a) PushT



(b) TwoRoom



(c) Reacher



(d) Cube

Figure 5: Perturbation visualizations on the four tasks. Each panel is a 2×4 grid: additive Gaussian pixel noise on row 1 (Original, $\sigma \in \{0.03, 0.05, 0.08\}$) and Gaussian blur on row 2 ($k \in \{3, 7, 11, 15\}$, auto sigma). The $\sigma = 0.08$ endpoint is the primary Gaussian-noise stress condition used in Tables 1 and 8.

B.10 Artifact and key-name mapping

The main text uses reader-facing diagnostic names. Exact relative artifact paths and implementation keys are collected here so that the manuscript remains readable while preserving reproducibility.

Table 10: Mapping from paper names to released relative paths and implementation keys.

Paper name	Released relative path or key
training-noise ceiling $\sigma_{\max}$	`std_max`
canonical evaluation artifact	`assets/paper1_data/canonical_evals_20260517.json`
canonical diagnostic artifact	`assets/paper1_data/canonical_diagnostics_20260517.json`
partial-correlation bootstrap artifact	`assets/paper1_data/partial_corr_bootstrap_20260523.json`
LeWM ACPC-basin artifact	`assets/paper1_data/acpc_basin_diagnostics.json`
PLDM ACPC-basin artifact	`assets/paper1_data/acpc_basin_diagnostics_pldm.json`
Phase-0 paired ACPC artifact	`assets/paper1_data/acpc_phase0_diagnostics.json`
blur baseline artifact	`assets/paper1_data/canonical_blur_baselines_20260523.json`
target-view ablation artifact	`assets/paper1_data/target_view_closed_loop_summary.json`
PLDM full-diagnostic artifact	`assets/paper1_data/canonical_full_diagnostics_pldm_20260523.json`
encoder-shift ratio	`noise_to_nn_cos_ratio`
fragility ratio	`predictor_target_to_nn_cos_ratio_at_max_std`
transition-resolution ratio	`transition_resolution_ratio_{cos,l2}`
multi-step rollout drift	`predictor_rollout_T8_l2`
ID probe R^2	`id_probe_r2`
main plot script	`tools/paper1_figs.py`
ACPC basin script	`tools/paper1_acpc_basin.py`
selective-contraction rendering script	`tools/paper1_selective_contraction.py`

C Diagnostic framework definitions

C.1 LeWM training objective

LeWM [26] is trained with latent prediction and SIGReg regularization:

$$\mathcal{L}_{\text{LeWM}} \;=\; \mathcal{L}_{\text{pred}} \;+\; \lambda \cdot \mathcal{L}_{\text{SIGReg}}\,, \tag{9}$$

where $\mathcal{L}_{\text{pred}}$ is the latent-space MSE between predicted and target representations. SIGReg projects each latent onto unit-norm random directions, computes the Epps–Pulley empirical-characteristic-function distance [8] between each projection and $\mathcal{N}(0, 1)$, and aggregates with Gauss-window weights to prevent collapse without explicit BatchNorm. Inference uses CEM [22] for latent-space MPC [9].

C.2 Five-layer diagnostic framework

The diagnostic framework decomposes the JEPA world-model control pass into pixels $\rightarrow$ encoder f $\rightarrow$ latent z $\rightarrow$ predictor g $\rightarrow$ cost surface $\rightarrow$ planned actions. Layers 1–2 describe whether visual perturbations enter the latent and how they change latent geometry. Layer 3 measures how the predictor amplifies an encoder shift. Layer 4 injects noise directly into z to separate encoder effects from predictor and cost effects. Layer 5 checks task-relevant information retained in the latent. We use nearest-neighbor distance as a local latent scale primitive in the spirit of kNN out-of-distribution detection [34]; the scale-normalized ratios are introduced for this diagnostic study.

Minimal notation. Let $z_i = f(x_i)$ and $\tilde{z}_i = f(\tilde{x}_i)$ be unperturbed and noised latents for the same token. The distance, local nearest-neighbor (NN) scale, and three introduced ratios are

$$\begin{aligned}
d_{\cos}(a,b) &= 1 - \frac{a^\top b}{\|a\|\,\|b\|}, \\
d_i^{\mathrm{NN}} &= \min_{j\neq i} d_{\cos}(z_i, z_j), \\
r_i^{\mathrm{enc}} &= \frac{d_{\cos}(z_i, \tilde{z}_i)}{d_i^{\mathrm{NN}} + \epsilon}, \\
r_i^{\mathrm{pred}} &= \frac{d_{\cos}\big(g(z)_i, g(\tilde{z})_i\big)}{d_i^{\mathrm{NN}} + \epsilon}, \\
R_d &= \frac{\mathrm{median}_t\ d(z_t, z_{t+1})}{\mathrm{median}_{(t,t')\in\mathcal{F}}\ d(z_t, z_{t'}) + \epsilon}.
\end{aligned} \tag{10}$$

Here r_i^{enc} is the encoder-shift ratio, r_i^{pred} is the fragility ratio, and R_d is the transition-resolution ratio for either cosine or L2 distance d. $\mathcal{F}$ denotes random far pairs from different temporal locations, and $\epsilon = 10^{-8}$ is a numerical-stability constant.

Layer metrics. Layer 1 reports raw angular/L2 encoder shift, angular gain per unit noise, and the encoder-shift ratio in Equation (10). Layer 2 reports local neighborhood scale, effective rank [30], and CKA [20]. Layer 3 reports fragility ratio and multi-step rollout L2 drift. Layer 4 reports latent-noise cost-surface slope and latent robust radius. Layer 5 reports the L2/cosine transition-resolution ratio and inverse-dynamics linear-probe R^2 [1]. Layers 1, 2, 4, and 5 are descriptive; Layer 3 supplies the two full-coverage cross-checkpoint metrics used in Section 5.5.

C.3 Cross-checkpoint validation protocol

For each task we compute Spearman ρ across the 9 LeWM checkpoints and partial Spearman conditioned on $\sigma_{\max}$. The partial step removes the sweep-level training-noise trend. We report 95% percentile bootstrap intervals ($B = 1000$, seed $= 42$) in the released artifact listed in Appendix B.10.

D Full diagnostic profile of LeWM-base on four tasks

Table 11 summarizes every core diagnostic on the four LeWM-base checkpoints. Exact released keys and diagnostic families are mapped in Appendix B.10; the table itself uses reader-facing metric names.

Table 11: Full LeWM-base diagnostics across four tasks.

Layer / metric	TwoRoom	PushT	Reacher	Cube	Unit / note
Encoder geometry					
Local NN cosine distance	0.0449	0.2360	0.0633	0.1856	cosine distance
Pairwise cosine distance	0.9904	1.0228	1.0252	1.0193	pair-wise cos dist
Effective rank	47.60	76.42	61.04	73.25	effective rank
Noise sensitivity					
Median noise angle (@ std 0.05)	5.51°	1.33°	3.22°	1.40°	median angular shift
Encoder-shift ratio	0.1031	0.011	0.0249	0.016	noise/NN cos ratio
Robust radius	0.0142	0.0537	0.0142	0.0356	critical noise std
First high-risk std	>0.08	>0.08	>0.08	>0.08	first high-risk std
Angular gain per std	1085.8	284.8	831.7	327.0	°/std angular gain
Geometry label	balanced	robust	balanced	robust	geometry label
Task resolution					
transition-resolution ratio (L2)	0.7216	0.3015	0.3704	0.4847	L2 resolution ratio
transition-resolution ratio (cos)	0.5538	0.0868	0.1351	0.2347	cos resolution ratio
ID probe R^2	0.2889	0.7739	0.1621	0.6657	linear probe R^2
Minimum ID probe R^2	0.2599	0.6786	0.1366	0.0972	min probe R^2
LiDAR rank proxy	46.06	13.95	45.90	42.46	LiDAR rank proxy
Action effect					
Mean action-induced prediction shift	0.5329	0.1283	0.2518	0.2364	action-perturb mean shift
Action-shift correlation	0.2847	0.2873	0.4042	0.2559	shift $\times$ $\|a\|$ corr
Predictor stability					
target/NN ratio	1.51e-4	3.54e-6	2.67e-5	3.39e-6	max-std target/NN ratio
Multi-step rollout drift	18.62	18.65	15.17	20.20	T=8 rollout L2 drift
Latent noise					
Linear CKA at max std	0.1986	0.5536	0.3085	0.1814	CKA unperturbed vs noisy
Latent cost-surface slope	635.31	1.3886	599.45	0.6208	goal latent cost slope

E Unconditioned sweep-trend correlations

The main text uses partial Spearman conditioned on $\sigma_{\max}$ as the residual cross-checkpoint diagnostic. Table 12 reports the unconditioned LeWM $n = 9$ trends only for transparency: both the diagnostics and the noise drop move strongly with the upstream training-noise level, so these rows should not be read as metric-as-oracle evidence.

Table 12: Sweep-trend correlations before conditioning on training-noise level: unconditioned LeWM $n = 9$ per-task Pearson r / Spearman ρ versus noise drop (unperturbed − observation-noise 0.08, unperturbed goal). Eval values come from the unified 3-seed × 100 protocol; Table 5 is the main residual diagnostic check.

Metric	TwoRoom (r/ρ)	PushT (r/ρ)	Reacher (r/ρ)	Cube (r/ρ)
fragility ratio	+0.92/+0.20	−0.55/−0.80	+0.98/+0.55	+0.39/+0.27
rollout drift	+0.81/+0.31	+0.98/+0.93	+0.99/+0.58	+0.91/+0.48

F PushT detailed partial-correlation view

Table 13 gives the detailed PushT $n = 9$ view for fragility ratio. It separates residual checkpoint quality from gap prediction. Lower fragility ratio aligns with better unperturbed and noisy success after conditioning on $\sigma_{\max}$, but the apparent correlation with the noise drop is mediated by $\sigma_{\max}$.

Table 13: Spearman ρ (unconditional) and partial Spearman ρ conditioned on $\sigma_{\max}$, for fragility ratio on the PushT $n = 9$ LeWM sweep under the unified 3-seed × 100 eval. CIs are percentile 95% intervals over $B = 1000$ bootstrap resamples (seed = 42). The $\sigma_{\max}$-only top rows are univariate sweep diagnostics with no metric/outcome pairing, so no CI is reported.

Quantity	Spearman ρ	95% bootstrap CI
$\rho(\sigma_{\max}, \text{metric})$	+0.83	—
$\rho(\sigma_{\max}, \text{unperturbed})$	0.00	—
$\rho(\sigma_{\max}, \text{obs-noise } 0.08)$	+0.88	—
$\rho(\sigma_{\max}, \text{noise drop})$	−0.98	—
$\rho(\text{metric}, \text{unperturbed})$ unconditional	−0.33	$[-0.95, +0.62]$
$\rho(\text{metric}, \text{unperturbed}) \mid \sigma_{\max}$ partial	−0.59	$[-0.97, -0.10]$
$\rho(\text{metric}, \text{obs-noise } 0.08)$ unconditional	+0.60	$[-0.11, +1.00]$
$\rho(\text{metric}, \text{obs-noise } 0.08) \mid \sigma_{\max}$ partial	−0.53	$[-0.84, 0.00]$
$\rho(\text{metric}, \text{noise drop})$ unconditional	−0.80	$[-1.00, -0.23]$
$\rho(\text{metric}, \text{noise drop}) \mid \sigma_{\max}$ partial	+0.19	$[0.00, +0.70]$

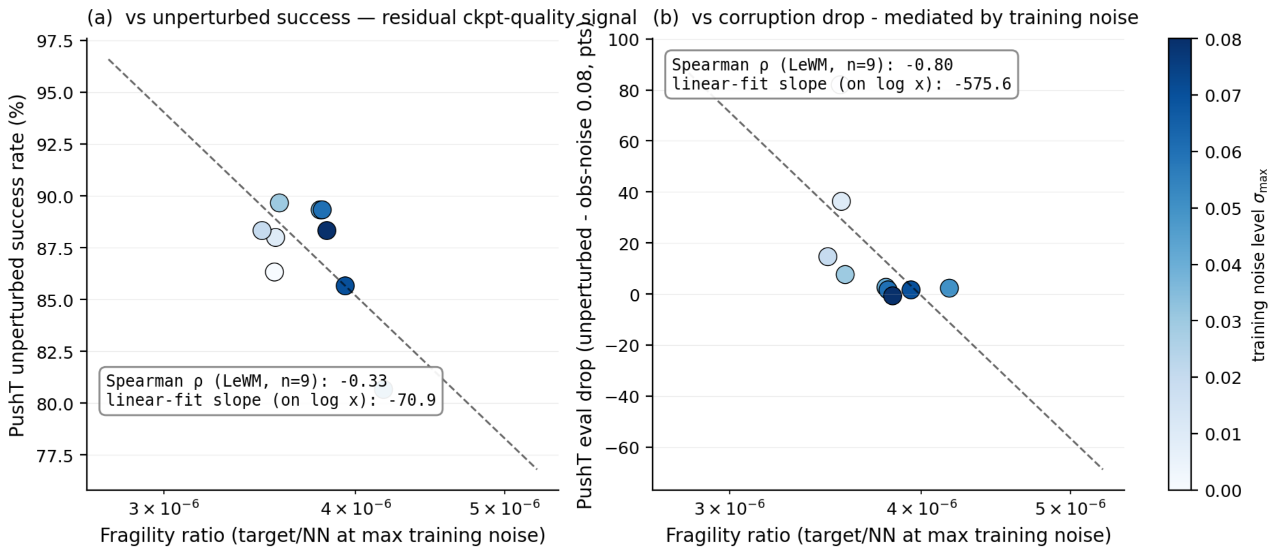


Figure 6: PushT $n = 9$ LeWM sweep: fragility ratio is a residual checkpoint-quality signal for unperturbed success (a); the apparent noise-drop correlation in (b) is mediated by $\sigma_{\max}$ (encoded by the color bar).

Panel (a) plots fragility ratio against unperturbed success. Panel (b) plots fragility ratio against noise drop; the color bar shows that low-$\sigma_{\max}$ checkpoints have high drop, high-$\sigma_{\max}$ checkpoints have low drop, and the metric tracks $\sigma_{\max}$ itself. After the $\sigma_{\max}$ trend is removed, the metric does not robustly separate small-drop from large-drop checkpoints.

G Heteroscedastic-loss formulation

The scale-preserving heteroscedastic NLL referenced in Section 6 and detailed in Section H is:

$$\mathcal{L}_{\text{hetero}} = \tfrac{1}{2}\,\exp(-s_t)\,\|z_{t+1} - \hat{z}_{t+1}\|^2 + \tfrac{1}{2}\,s_t\,, \tag{11}$$

where s_t is the σ-head-predicted log-variance. During training $\exp(-s_t)$ acts as an automatic weight: high-error transitions are down-weighted and low-error transitions are up-weighted. When $s_t \equiv 0$ the loss reduces to plain MSE.

H Heteroscedastic-loss negative result (data)

This appendix records the full heteroscedastic-loss data referenced in Section 6. The σ-head is trained jointly with the predictor; gradient is detached on the σ path so the mean prediction path is exactly LeWM MSE when σ is constant.

Table 14: Heteroscedastic-loss evaluation (canonical 3-seed $\times$ 100 protocol). The LeWM+noise rows use representative observation+goal 0.08 high-noise rows (TwoRoom $\sigma_{\max} = 0.08$, PushT $\sigma_{\max} = 0.06$), matching the strongest stress column of this ablation; the observation-only representative rows in Table 8 differ.

Task / model	unpert.	goal .05	obs .05	obs+goal .05	goal .08	obs+goal .08
TwoRoom LeWM-base	94.00	73.33	72.33	61.33	58.67	50.00
TwoRoom LeWM+noise row	98.33	98.00	98.33	98.00	98.67	98.67
TwoRoom hetero	99.67	85.33	96.67	84.67	73.33	55.33
PushT LeWM-base	86.33	38.00	17.00	12.00	11.33	4.67
PushT LeWM+noise row	89.33	87.67	88.00	88.33	89.67	87.00
PushT hetero	13.33	7.67	7.67	7.67	9.67	6.00

TwoRoom hetero reaches 99.67% on unperturbed evaluation — compatible with the prior that low-dimensional discrete tasks can tolerate stronger nuisance contraction / clustering — but lags noise training on high-noise robustness. **PushT hetero unperturbed success is** 13.33% — **a method-level failure**, not evidence for a useful robustness compromise.

Table 15: Representation diagnostics of the hetero-loss failure.

Metric	TwoRoom base	TwoRoom hetero	PushT base	PushT hetero
Local NN cosine distance	0.0449	0.0281	0.2360	0.1051
Effective rank	47.60	33.59	76.42	42.85
transition-resolution ratio (cos)	0.5538	0.3780	0.0868	0.0101
transition-resolution ratio (L2)	0.7216	0.6055	0.3015	0.1023
ID probe R^2	0.2889	-0.0573	0.7739	0.2678
Mean action-induced prediction shift	0.5329	0.4482	0.1283	0.0841
Multi-step rollout drift	18.62	17.90	18.65	14.01

Hetero-loss compresses the representation in both tasks. In TwoRoom (low-dimensional, discrete, redundant), the observed compression coexists with high unperturbed success, but the robustness shortfall shows that compression alone is not the target. For PushT, the L2 transition-resolution ratio collapses from 0.3015 to 0.1023 and ID probe R^2 from 0.7739 to 0.2678 — task-relevant state information is erased. The contrast with input-side noise training is sharp: the fixed $\sigma_{\max} = 0.08$ PushT checkpoint in Table 4 leaves rank and probe nearly unchanged, so this collapse is specific to error-based downweighting. The drop in multi-step rollout drift is not good news either: the latent has become more *predictable* without being more *controllable*.

Mechanism. The learned σ values align with visible high-error transition regimes (contact frames in PushT, doorway crossings in TwoRoom, etc.), but using σ as an automatic weight to down-weight high-error transitions misclassifies the contact-and-control-critical states of PushT as "unimportant" and erases them. This is a negative example of the broader selective-consistency lesson: in contact-heavy control, **hard $\neq$ unimportant**.

This finding motivates a narrower follow-up direction: per-token *consistency* controllers should be separated from loss reweighting, and any difficulty signal should be detached from the mean prediction path unless it is explicitly protected by an action-relevance guard. We leave that objective design outside the present diagnostic study.

I Mechanism-attribution probe

Section 5.4 reports what moves in the representation and Section 5.5 reports which diagnostics correlate with evaluation across checkpoints. To localize whether the perturbation enters through the encoder or later predictor/planner stages, we use latent-noise probing as a conservative mechanism check. Directly injecting noise into latent z skips the encoder and decouples encoder contributions from predictor and cost contributions.

Table 16: Three-layer attribution by task (LeWM $n = 9$ sweep, unified 3-seed $\times$ 100 eval).

Task	Strongest signal in these probes	Residual after removing the $\sigma_{\max}$ trend
TwoRoom	encoder-shift signal (rank-saturated)	near-zero residual; saturation limits inference
PushT	encoder + single-step predictor (both mediated by $\sigma_{\max}$)	no residual metric isolates noise drop
Reacher	encoder + multi-step rollout	weak multi-step drift residual signal (partial $\rho = +0.37$)
Cube	encoder	weak residual on multi-step drift (partial $\rho = +0.23$)

Across these probes, the strongest common signal is encoder-side perturbation transduced by the predictor. PushT is the clearest mediated case: multi-step rollout drift has unconditional $\rho = +0.93$ against noise drop, but partial $\rho = -0.01$ after removing the $\sigma_{\max}$ trend; the single-step fragility ratio similarly changes from unconditional $\rho = -0.80$ to partial $\rho = +0.19$. Reacher and Cube retain only weak residual multi-step signals, and TwoRoom is limited by saturation. A stronger causal claim about planner or cost-surface contributions would require multi-task latent-injection or cost-ranking ablations.

J Cross-method replication: PLDM noise sweep on four tasks

This appendix records the second method-family on which the noise sweep and the cross-checkpoint correlation analysis replicate. PLDM follows the joint-embedding predictive line from Sobal et al. [32]; the concrete latent-dynamics planning baseline is from Sobal et al. [33]. We use the implementation distributed through the `stable-worldmodel` baseline suite [25]. Training and evaluation follow the LeWM canonical protocol bit-for-bit: per-frame Gaussian pixel noise, $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$, three evaluation seeds (42/43/44), 100 trajectories per seed.

Coverage. The PLDM release is complete for the same grid as the LeWM canonical sweep: 4 tasks $\times$ 9 configurations (no-noise baseline plus $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$), three evaluation seeds (42/43/44), and 100 trajectories per seed. The PLDM eval, predictor-diagnostic, full-diagnostic, and cross-method-correlation JSON files are listed in the data manifest; we do *not* merge them into the LeWM canonical files used by Tables 1–13.

No-noise-trained PLDM cliff. PLDM reproduces the no-noise-trained Gaussian-noise cliff on PushT and TwoRoom, while Reacher and Cube show much smaller drops under this PLDM training recipe (Table 17). This supports a method-family reading of the failure mode without claiming every task has the same cliff magnitude.

Table 17: PLDM baseline trained without input-noise augmentation on all four tasks. Success rate mean $\pm$ population std, 3 evaluation seeds $\times$ 100 trajectories. Drop is clean success minus observation-noise 0.08 success.

Task	unpert.	obs 0.05	obs 0.08	obs+goal 0.08	drop
TwoRoom	96.67 ± 1.70	79.67 ± 4.03	67.00 ± 2.16	51.67 ± 2.05	29.67
PushT	75.33 ± 3.68	46.00 ± 4.97	18.33 ± 2.05	10.00 ± 2.16	57.00
Reacher	82.67 ± 1.70	81.33 ± 4.03	80.33 ± 5.73	79.33 ± 0.94	2.33
Cube	52.67 ± 3.30	50.33 ± 4.50	48.33 ± 4.50	48.33 ± 2.87	4.33

PLDM full sweep — unperturbed evaluation. Table 18 reports the per-task unperturbed success rate at every trained $\sigma_{\max}$ level. Table 19 reports the corresponding observation-noise 0.08 robustness with an unperturbed goal.

Table 18: PLDM+noise sweep, unperturbed success rate (%). † marks one representative high grid row per task for compact reading.

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	96.67 ± 1.70	75.33 ± 3.68	82.67 ± 1.70	52.67 ± 3.30
0.01	96.33 ± 0.94	72.33 ± 4.19	78.00 ± 0.82	51.33 ± 1.25
0.02	97.33 ± 0.47	71.33 ± 3.86	82.33 ± 2.36	53.33 ± 1.70
0.03	96.67 ± 1.70	$76.67 \pm 7.54^{\dagger}$	83.00 ± 3.74	52.67 ± 3.30
0.04	97.67 ± 0.47	68.67 ± 5.25	$85.00 \pm 2.16^{\dagger}$	54.00 ± 2.94
0.05	97.67 ± 1.25	74.33 ± 3.40	72.00 ± 1.41	54.00 ± 2.16
0.06	98.00 ± 0.82	70.33 ± 6.18	79.00 ± 0.82	$56.00 \pm 4.97^{\dagger}$
0.07	98.00 ± 0.82	66.67 ± 8.38	80.67 ± 2.62	53.67 ± 3.68
0.08	$98.33 \pm 0.94^{\dagger}$	68.00 ± 1.41	80.33 ± 1.25	54.00 ± 1.63

Table 19: PLDM+noise sweep, observation-noise 0.08 success rate with unperturbed goal (%). ‡ marks one representative high grid row per task for compact reading.

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	67.00 ± 2.16	18.33 ± 2.05	80.33 ± 5.73	48.33 ± 4.50
0.01	96.00 ± 1.63	23.33 ± 2.49	77.67 ± 3.40	51.33 ± 2.36
0.02	95.67 ± 0.94	47.00 ± 0.82	80.67 ± 5.25	51.00 ± 4.32
0.03	96.33 ± 1.25	$73.00 \pm 7.48^{\ddagger}$	$81.67 \pm 4.71^{\ddagger}$	54.33 ± 2.62
0.04	97.33 ± 0.94	66.67 ± 5.73	80.33 ± 2.87	$54.67 \pm 2.36^{\ddagger}$
0.05	97.67 ± 1.89	71.67 ± 5.79	62.67 ± 4.19	54.00 ± 3.74
0.06	$98.00 \pm 0.00^{\ddagger}$	69.33 ± 4.78	75.00 ± 0.82	54.33 ± 0.94
0.07	98.00 ± 0.82	64.00 ± 7.79	78.33 ± 2.05	53.00 ± 3.74
0.08	97.67 ± 1.25	68.00 ± 5.66	79.33 ± 1.25	53.67 ± 2.05

Reading. Four quantitative observations are useful for interpreting PLDM as a second method family:

1. **TwoRoom (visually redundant) shows large recovery on both methods.** PLDM observation-noise 0.08 success rises from 67.00% at the no-noise baseline to 98.00% at $\sigma_{\max} = 0.06$. This mirrors the LeWM reading that TwoRoom tolerates strong same-state nuisance contraction.
2. **PushT (contact-heavy) exhibits the Gaussian-noise cliff and large noise-training recovery.** PLDM without noise augmentation drops by 57.00 pt under observation-noise 0.08; training with $\sigma_{\max} = 0.03$ recovers observation-noise 0.08 success to 73.00% (+54.67 pt over the no-noise baseline). The unperturbed and observation-noise 0.08 representative high rows are co-located at $\sigma_{\max} = 0.03$ on PLDM, giving a compact grid-level reference for the recovery pattern.
3. **Reacher is already robust under no-noise-trained PLDM.** The no-noise PLDM drop is only 2.33 pt, and a representative high observation-noise row occurs at $\sigma_{\max} = 0.03$ (81.67%). This makes Reacher a low-cliff external baseline for interpreting noise-training gains.
4. **Cube (structured 3D manipulation) remains weakly noise-responsive.** PLDM Cube unperturbed success spans 51.33–56.00% and observation-noise 0.08 spans 48.33–54.67% across the full grid. This is the same weak noise responsiveness seen on LeWM.

Method-specific details. PLDM's representative PushT unperturbed row (76.67% at $\sigma_{\max} = 0.03$) is lower than LeWM's corresponding representative row (89.67% at $\sigma_{\max} = 0.03$). The external baseline is therefore a robustness check: the cliff-and-recovery shape is reproduced, while the exact high-grid location is method-dependent. The LeWM PushT split between unperturbed and noisy representative rows is a LeWM-family signature within this fixed protocol.

Cross-method partial correlations. We repeat the fragility ratio partial-correlation analysis for PLDM on all four tasks. The headline PushT null replicates in the limited sense that we do not see stable evidence for a non-zero residual Gaussian-noise-gap association: PLDM has unconditional $\rho(\text{metric}, \text{noise drop}) = -0.75$ but partial $\rho(\text{metric}, \text{noise drop}) \mid \sigma_{\max} = -0.05$ (95% bootstrap CI $[-0.92, +0.61]$), and the joint LeWM+PLDM PushT partial after conditioning on $\sigma_{\max}$ and method is +0.22 (95% CI $[-0.59, +0.61]$). The intervals are wide and include zero. Other tasks show task-specific residuals, so the full table is evidence for a bounded diagnostic reading rather than a general robustness predictor.

Table 20: PLDM fragility ratio partial correlations, $n = 9$ per task. The first three partial columns condition on $\sigma_{\max}$ within PLDM; the final column uses the joint LeWM+PLDM sample ($n = 18$) and conditions on both $\sigma_{\max}$ and a method dummy.

Task	n	unpert.	obs 0.08	drop	joint drop
TwoRoom	9	−0.26	−0.43	+0.35	+0.12
PushT	9	−0.22	−0.13	−0.05	+0.22
Reacher	9	−0.68	−0.49	+0.17	+0.43
Cube	9	−0.36	+0.19	−0.53	−0.13

PLDM full-sweep ACPC basin replication. We also run the Gaussian-noise basin protocol on all 36 PLDM checkpoints. Table 21 reports the compact baseline-vs-representative observation-noise summary; the full

4 tasks × 9 configurations grid is listed in Appendix B.10.

Table 21: PLDM full-sweep Gaussian-noise ACPC basin replication, summarized by the no-noise baseline and each task's representative high-noise PLDM observation-noise 0.08 grid row. The protocol matches Table 3: same-state Gaussian-noise views of the observation history (std 0.01–0.08), unperturbed goal, and the same recorded action sequence.

Task	checkpoint	obs 0.08	drop	R_E	R_F
TwoRoom	base	67.00	29.67	3.287	0.611
TwoRoom	noise 0.06	98.00	0.00	0.235	0.034
PushT	base	18.33	57.00	0.874	0.846
PushT	noise 0.03	73.00	3.67	0.153	0.091
Reacher	base	80.33	2.33	0.237	0.060
Reacher	noise 0.03	81.67	1.33	0.135	0.039
Cube	base	48.33	4.33	0.831	0.453
Cube	noise 0.04	54.67	−0.67	0.146	0.054

Reading. The PLDM basin replication is directionally consistent with the LeWM ACPC-basin reading: the reported representative observation-noise 0.08 checkpoint reduces R_F on all four tasks, sharply on TwoRoom, PushT, and Cube. Reacher is the boundary case: PLDM is already near-robust at the no-noise baseline, and both the behavioral drop and the basin change are small. The full 36-row artifact reduces the concern that the paired basin pattern is purely a LeWM artifact, but it does not upgrade the claim to a general mechanism because Table 22 shows different internal diagnostic movement.

PLDM full-diagnostic mechanism check. We also run the same five diagnostic layers on the PLDM sweep. The compact PLDM base-vs-representative view in Table 22 matches the dominant fixed-0.08 LeWM movement in Table 4: both families preserve rank, transition resolution, and the inverse-dynamics probe to first order while strongly reducing multi-step predictor drift. The clearest method-specific difference is TwoRoom: LeWM's fixed-0.08 checkpoint also compresses effective rank (47.6 → 37.7), whereas PLDM's representative row leaves rank essentially unchanged (74.8 → 72.7). This separates two claims: the task-level fragility/recovery signature replicates across method families, while the residual diagnostic movement remains architecture-dependent.

Table 22: PLDM full-diagnostic check, no-noise baseline → representative noise-trained checkpoint. Representatives are high-noise PLDM observation-noise 0.08 grid rows (TwoRoom $\sigma_{\max} = 0.06$, PushT $\sigma_{\max} = 0.03$, Reacher $\sigma_{\max} = 0.03$, Cube $\sigma_{\max} = 0.04$).

Task	rep. $\sigma_{\max}$	eff. rank	trans. L2	ID probe R^2	rollout T8 L2
TwoRoom	0.06	74.78 → 72.70	0.8188 → 0.8254	0.3233 → 0.3201	20.36 → 1.02
PushT	0.03	130.13 → 131.90	0.4710 → 0.4778	0.7570 → 0.7479	20.25 → 2.82
Reacher	0.03	117.52 → 108.14	0.5053 → 0.4824	0.2150 → 0.1844	1.45 → 0.94
Cube	0.04	134.80 → 124.49	0.6395 → 0.6313	0.5428 → 0.5576	18.98 → 4.02

What this strengthens, and what remains bounded. PLDM reduces the concern that the four task signatures are tied to one LeWM checkpoint family. The full PLDM basin artifact, summarized in Table 21, further supports the paired same-state perturbation reading at the baseline-vs-representative high-noise grid rows. It also supports the diagnostic-boundary claim: after removing the sweep-level $\sigma_{\max}$ effect and the method offset on PushT, the local predictor-sensitivity diagnostic does not isolate noise-specific robustness. The TwoRoom and Cube PLDM residuals remain visible in Table 20. The intervals are wide and the noisy-eval ranges are small on the saturated tasks, so the residuals remain bounded diagnostics.

Mechanism boundary. Table 22 is the reason we do not promote any single internal mechanism to a general explanation. On both families the most consistent representative-checkpoint movement is a strong reduction in multi-step rollout drift, but the residual movement is method-specific: LeWM compresses TwoRoom's effective rank where PLDM does not, and the two families place their weak residual partial-correlation signals on different tasks (Tables 5 and 20). The shared conclusion is therefore about *control-time visual fragility and noise-training recovery*; the internal diagnostic route is method-specific.

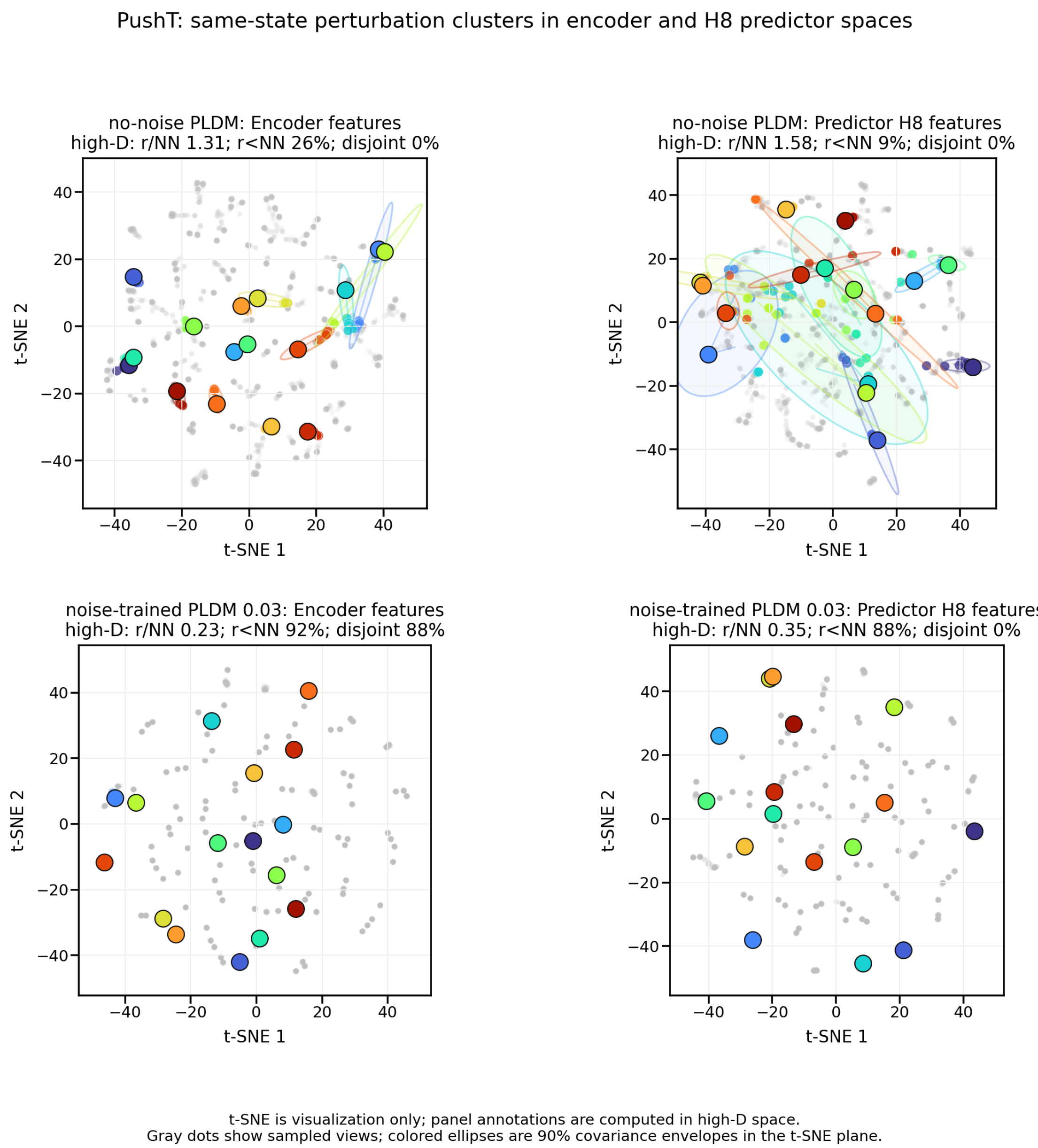


Figure 7: Qualitative PLDM PushT same-state perturbation clusters for Table 21. Top: no-noise-trained PLDM; bottom: noise-trained PLDM at $\sigma_{\max} = 0.03$. Small panel summaries report original-space local statistics; t-SNE coordinates and envelopes are qualitative visualization only and not comparable with LeWM.

K Cross-perturbation sanity check: no-noise-trained blur evaluation

This appendix addresses the narrow concern that the observed visual fragility is an artefact of Gaussian noise only. We evaluate the LeWM and PLDM baselines trained without input-noise augmentation under Gaussian blur with kernel sizes $3, 7, 11, 15$, applied to pixels, goal images, or both. This is an **eval-only** stress test: no blur-trained checkpoint is included, and the blur data are not mixed into the Gaussian-noise sweep. The released source is listed in Appendix B.10.

Table 23: Observation+goal blur stress test on baselines trained without input-noise augmentation. Success rate mean ± population std over 3 evaluation seeds × 100 trajectories. Kernel sizes 11 and 15 are severe low-pass endpoints on 224×224 inputs, so they should not be read as mild perturbations.

Method	Task	unpert.	$k = 3$	$k = 7$	$k = 11$	$k = 15$	worst drop
LeWM	TwoRoom	94.00 ± 3.56	39.33 ± 2.05	32.67 ± 3.09	30.33 ± 2.87	30.67 ± 0.47	63.67
LeWM	PushT	86.33 ± 2.36	81.67 ± 4.50	74.00 ± 1.41	65.33 ± 2.49	58.67 ± 6.65	27.67
LeWM	Reacher	58.67 ± 1.25	57.00 ± 6.38	45.67 ± 2.05	36.00 ± 4.90	20.33 ± 2.49	38.33
LeWM	Cube	66.67 ± 2.62	65.00 ± 1.63	62.33 ± 2.62	57.33 ± 2.87	54.00 ± 2.16	12.67
PLDM	TwoRoom	96.67 ± 1.70	38.33 ± 0.47	30.33 ± 2.87	28.33 ± 1.25	28.33 ± 1.70	68.33
PLDM	PushT	75.33 ± 3.68	74.00 ± 2.94	72.00 ± 3.56	67.67 ± 4.03	62.33 ± 2.05	13.00
PLDM	Reacher	82.67 ± 1.70	86.00 ± 2.16	80.00 ± 3.56	72.00 ± 2.16	68.00 ± 4.08	14.67
PLDM	Cube	52.67 ± 3.30	53.00 ± 5.35	53.00 ± 2.83	50.67 ± 3.68	52.33 ± 5.44	2.00

Reading. Blur is a different failure mode from Gaussian pixel noise. The clear collapse is TwoRoom: it falls below 40% already at $k = 3$ on both LeWM and PLDM and then saturates around 28–33% for $k = 7$–15. This reverses the Gaussian-noise ordering, where PushT is the most fragile task. The other tasks are more stable under blur: PLDM PushT, PLDM Reacher, and both Cube baselines lose at most about 15 pt at the strongest endpoint, while LeWM PushT degrades gradually and LeWM Reacher drops sharply only at $k = 15$. We therefore use blur only as an eval-only cross-perturbation sanity check: visual fragility is not unique to Gaussian noise, but the task ordering and recovery profile are perturbation-specific.

L Phase-0 paired ACPC diagnostics

L.1 Exploratory readout definitions

The main text fixes ACPC to the $H = 8$ identity-readout rollout-latent basin diagnostic. The additional readouts below are exploratory and are not used for model selection.

Encoder perturbation difference (EPD).
$\mathrm{EPD} = d(E(o), E(\tilde{o}))$. This reports whether a perturbation enters the latent, not whether it matters for control.

One-step consistency (ACPC-1).
The one-step paired score is

$$\mathrm{ACPC}_1 = d\big(F(E(o), a), F(E(\tilde{o}), a)\big).$$

For cross-checkpoint comparisons we optionally divide by the natural transition magnitude $d(z_t, z_{t+1}) + \epsilon$.

Multi-step consistency (ACPC-H).
Equation (3) with $H \in \{4, 8\}$ measures disagreement between unperturbed and perturbed rollouts under the same action sequence.

Predictive cost consistency (PCC).
$\mathrm{PCC} = \big|J(F^H(E(o), \mathbf{a}), g) - J(F^H(E(\tilde{o}), \mathbf{a}), g)\big|$ for a cost/goal readout J and goal g. PCC tests whether predictive mismatch changes the planning cost.

Candidate-ranking agreement (CRA).
For a shared candidate set $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$, CRA is the Spearman/Kendall agreement of the unperturbed and perturbed cost vectors.

Margin-conditioned action flip (MAF).
$\mathrm{MAF} = \mathbf{1}\big[u^*_{\mathrm{orig}} \neq u^*_{\mathrm{noise}},\ C_{(2)} - C_{(1)} > \delta\big]$: a flip of the selected action counts only when the unperturbed top-1/top-2 cost margin exceeds δ.

Action-relevant discriminability margin (ADM).
Over action/transition-distinct pairs P_{disc}, $\mathrm{ADM} = \mathrm{median}_{(i,j)\in P_{\mathrm{disc}}}\, d(\Psi_i, \Psi_j)$. In this paper ADM is a proxy guard rather than a direct proof of the margin condition in Equation (6).

Selective predictive robustness ratio (SPRR).
A summary of action-distinct discriminability relative to original/perturbed predictive inconsistency,

$$\mathrm{SPRR} = \frac{\mathrm{median}_{(i,j)\in P_{\mathrm{disc}}}\, d(\Psi_i, \Psi_j)}{\mathrm{median}_{P_{\mathrm{pert}}}\, \mathrm{ACPC}_H + \epsilon}.$$

L.2 Phase-0 table

This appendix records the first full-coverage run of these paired action-conditioned diagnostics. The released artifact covers 72 checkpoint rows: LeWM and PLDM, four tasks, and nine training-noise levels per task. All rows completed successfully. Each row uses 100 sampled sequences, Gaussian observation+goal noise at std 0.08, rollout horizon $H = 8$, and a shared candidate-action set of one dataset future plus 64 random futures. Because this uses the auxiliary observation+goal stress endpoint rather than the paper's primary unperturbed-goal endpoint, it is reported as a broader exploratory sanity check rather than as the main ACPC evidence. PCC, CRA, and MAF are computed from the same original/noisy candidate-cost vectors. ADM and SPRR use an action-distance latent proxy, not an oracle state/keypoint margin.

Table 24: Phase-0 paired ACPC diagnostics, no-noise baseline → representative high observation+goal 0.08 checkpoint. Lower is better for ACPC-H/transition, PCC, and MAF; higher is better for CRA and top-8 overlap. Values are exploratory diagnostics.

Task	Method	robust $\sigma_{\max}$	obs+goal success	ACPC-H/trans.	PCC	CRA	top-8	MAF
TwoRoom	LeWM	0.08	50.00 → 98.67	1.254 → 0.042	176.6 → 7.0	0.157 → 0.990	0.201 → 0.952	0.92 → 0.06
TwoRoom	PLDM	0.05	51.67 → 98.33	1.062 → 0.063	164.3 → 7.1	0.367 → 0.982	0.287 → 0.911	0.85 → 0.05
PushT	LeWM	0.06	4.67 → 87.00	2.740 → 0.149	67.3 → 4.6	0.323 → 0.986	0.213 → 0.914	0.96 → 0.08
PushT	PLDM	0.03	10.00 → 72.00	1.682 → 0.162	52.0 → 8.1	0.300 → 0.967	0.286 → 0.869	0.90 → 0.08
Reacher	LeWM	0.02	15.00 → 85.67	1.922 → 0.023	528.0 → 1.2	0.242 → 0.999	0.219 → 0.990	0.97 → 0.01
Reacher	PLDM	0.04	79.33 → 81.33	0.104 → 0.063	11.7 → 6.3	0.965 → 0.987	0.930 → 0.949	0.13 → 0.10
Cube	LeWM	0.07	46.33 → 68.00	1.944 → 0.055	88.8 → 4.1	0.337 → 0.996	0.266 → 0.951	0.88 → 0.00
Cube	PLDM	0.08	48.33 → 56.33	1.136 → 0.034	113.6 → 3.0	0.663 → 0.997	0.512 → 0.967	0.74 → 0.01

Reading. The Phase-0 pass converts the ACPC definitions from a prospective protocol into a release artifact that behaves sensibly on the existing sweep. The largest Gaussian-noise cliffs (TwoRoom and PushT on both methods, Reacher/Cube on LeWM) coincide with large baseline ACPC-H, PCC, ranking disruption, and action flips, and the representative high observation+goal checkpoints reduce these quantities sharply. PLDM Reacher is the important boundary case: it already has high observation+goal success at the no-noise baseline, and the paired diagnostics are correspondingly already close to the recovered checkpoint. Across the joint LeWM+PLDM sample within each task, partial Spearman correlations after conditioning on $\sigma_{\max}$ and method remain task-specific: ACPC-H/transition vs. noise drop is strong on PushT (+0.67) and Reacher (+0.71), but only modest on TwoRoom (+0.21) and Cube (+0.33). We therefore treat Table 24 as a face-validity and local-diagnostic check, not as evidence that ACPC-H, PCC, CRA, MAF, ADM, or SPRR can by themselves predict robustness.

M Target-view ablation completion check

This appendix records the completed LeWM target-view ablation referenced in Section 6. The runs use the origin-future target configuration: the prediction context is encoded from the configured Gaussian-perturbed history, while the prediction target remains the original future latent. This is the perturbed-history → original-future branch, contrasted with the default full-sequence perturbed-target branch used by the main LeWM+noise sweep. The released summary artifact is listed in Appendix B.10.

We verified the four-task run set after completion: eight target-noise checkpoints per task and epoch-10 summaries for unperturbed, observation-noise 0.03, observation-noise 0.05, and observation-noise 0.08 evaluation. The canonical table below uses seeds 42/43/44 and 100 trajectories per seed.

Table 25: LeWM perturbed-history → original-future target-view ablation. Values are success rate mean ± population std over the eight target-noise checkpoints; each checkpoint value is the three-seed mean from `eval_summary.csv`.

Task	unperturbed	obs 0.03	obs 0.05	obs 0.08
TwoRoom	93.33 ± 1.72	88.67 ± 6.04	77.71 ± 9.12	61.75 ± 7.65
PushT	82.92 ± 4.00	57.71 ± 8.96	20.21 ± 11.68	6.75 ± 5.00
Reacher	59.92 ± 1.80	40.38 ± 2.78	32.00 ± 4.57	19.62 ± 3.97
Cube	65.00 ± 2.01	61.42 ± 2.06	44.29 ± 1.32	39.62 ± 1.22

Table 26: Closed-loop target-view comparison at observation-noise 0.08. Values are success rate mean ± population std over the eight training-noise checkpoints; each checkpoint value is the canonical three-seed mean.

Task	full-sequence perturbed target	perturbed-history → original future	full-seq advantage
TwoRoom	94.71 ± 1.90	61.75 ± 7.65	+32.96
PushT	72.83 ± 12.84	6.75 ± 5.00	+66.08
Reacher	76.17 ± 10.89	19.62 ± 3.97	+56.54
Cube	59.83 ± 5.55	39.62 ± 1.22	+20.21

Reading. The ablation acts as a negative scope check and supports retaining the full-sequence empirical branch. Simply changing the prediction target to the original future does not reproduce the recovery from input-side noise augmentation in Table 8: PushT and Reacher remain close to their no-noise robustness cliffs, TwoRoom degrades under stronger eval noise despite high unperturbed success, and Cube does not improve over the main sweep.

A matched PushT 0to008 check shows the contrast: rerunning both branches' $\sigma_{\max} = 0.08$ checkpoints under the same fixed evaluation code and canonical seeds gives observation-noise 0.08 three-seed means of 86.33% for the full-sequence checkpoint versus 8.67% for the origin-target checkpoint (the canonical sweep value for the same full-sequence checkpoint in Table 8 is 89.00%; the difference is normal evaluation variance). We interpret this as evidence consistent with a closed-loop train/eval consistency explanation: full-sequence noise augmentation better matches the planner's latent autoregressive feedback, while one-step origin-target denoising does not. The result leaves arbitrary history mismatches outside scope; the target view alone is not the missing mechanism, and an explicit action-conditioned predictive-consistency objective is still needed.